Gilbert Strang - Introduction to Linear Algebra
Chapters 1 - 7

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Contents

1 Chapter 1: Introduction to Vectors

1.1 Vectors and Linear Combinations

Problems 1-9 are about addition of vectors and linear combinations.

1. **Describe geometrically (line, plane, or all of R^3) all linear combinations of**

   (a) $\begin{bmatrix}1\\2\\3\end{bmatrix}$ *and* $\begin{bmatrix}3\\6\\9\end{bmatrix}$ **(b)** $\begin{bmatrix}1\\0\\0\end{bmatrix}$ *and* $\begin{bmatrix}0\\2\\3\end{bmatrix}$ **(c)** $\begin{bmatrix}2\\0\\0\end{bmatrix}$ *and* $\begin{bmatrix}0\\2\\2\end{bmatrix}$ *and* $\begin{bmatrix}2\\2\\3\end{bmatrix}$

2. **Draw $\boldsymbol{v} = \begin{bmatrix}4\\1\end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix}-2\\2\end{bmatrix}$ and $\boldsymbol{v}+\boldsymbol{w}$ and $\boldsymbol{v}-\boldsymbol{w}$ in a single xy plane.**

3. **If $\boldsymbol{v}+\boldsymbol{w} = \begin{bmatrix}5\\1\end{bmatrix}$ and $\boldsymbol{v}-\boldsymbol{w} = \begin{bmatrix}1\\5\end{bmatrix}$, compute and draw the vectors $\boldsymbol{v}$ and $\boldsymbol{w}$.**

4. **From $\boldsymbol{v} = \begin{bmatrix}2\\1\end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix}1\\2\end{bmatrix}$, find the components of $3\boldsymbol{v}+\boldsymbol{w}$ and $c\boldsymbol{v}+d\boldsymbol{w}$.**

5. **Compute $\boldsymbol{u}+\boldsymbol{v}+\boldsymbol{w}$ and $2\boldsymbol{u}+2\boldsymbol{v}+\boldsymbol{w}$. How do you know $\boldsymbol{u},\boldsymbol{v},\boldsymbol{w}$ lie in a plane?**

   These lie in a plane because $\boldsymbol{w} = c\boldsymbol{u}+d\boldsymbol{v}$. Find c and d

   $$\boldsymbol{u} = \begin{bmatrix}1\\2\\3\end{bmatrix}, \quad \boldsymbol{v} = \begin{bmatrix}-3\\1\\-2\end{bmatrix}, \quad \boldsymbol{w} = \begin{bmatrix}2\\-3\\-1\end{bmatrix}.$$

6. **Every combination of $\boldsymbol{v} = (1,-2,1)$ and $\boldsymbol{w} = (0,1,-1)$ has components that add to ______. Find c and d so that $c\boldsymbol{v}+d\boldsymbol{w} = (3,3,-6)$. Why is $(3,3,6)$ impossible?**

7. **In the xy plane mark all nine of these linear combinations:**

   $$c\begin{bmatrix}2\\1\end{bmatrix} + d\begin{bmatrix}0\\1\end{bmatrix} \quad \textbf{with} \quad c = 0,1,2 \quad \textbf{and} \quad d = 0,1,2.$$

8. **The parallelogram in Figure 1.1 has diagonal $\boldsymbol{v}+\boldsymbol{w}$. What is its other diagonal? What is the sum of the two diagonals? Draw that vector sum.**

9. **If three corners of a parallelogram are $(1,1)$, $(4,2)$, and $(1,3)$, what are all three of the possible fourth corners? Draw two of them.**

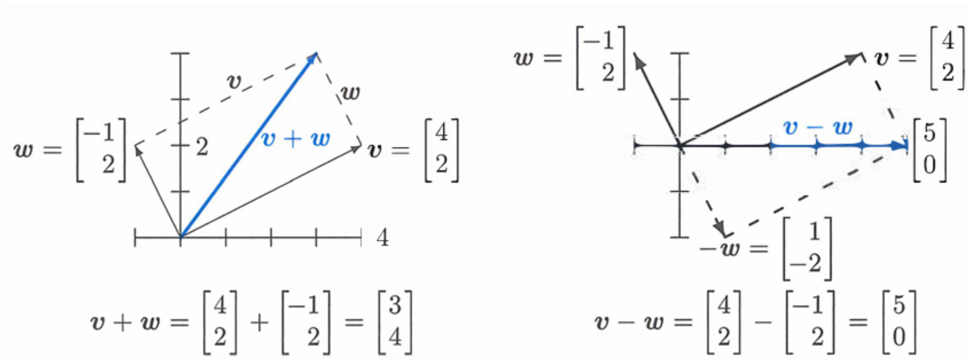


Figure 1.1: Vector addition $v+w = (3,4)$ produces the diagonal of a parallelogram. The reverse of w is $-w$. The linear combination on the right is $v - w = (5,0)$.

Problems 10–14 are about special vectors on cubes and clocks in Figure 1.4.

10. **Which point of the cube is $\boldsymbol{i}+\boldsymbol{j}$? Which point is the vector sum of $\boldsymbol{i} = (1,0,0)$ and $\boldsymbol{j} = (0,1,0)$ and $\boldsymbol{k} = (0,0,1)$? Describe all points (x,y,z) in the cube.**

11. **Four corners of this unit cube are $(0,0,0), (1,0,0), (0,1,0), (0,0,1)$. What are the other four corners? Find the coordinates of the center point of the cube. The center points of the six faces are ____________. The cube has how many edges?**

12. **Review Question. In xyz space, where is the plane of all linear combinations of $\boldsymbol{i} = (1,0,0)$ and $\boldsymbol{i} + \boldsymbol{j} = (1,1,0)$?**

13. **(a) What is the sum V of the twelve vectors that go from the center of a clock to the hours 1:00, 2:00, ..., 12:00?**

    (b) If the 2:00 vector is removed, why do the 11 remaining vectors add to 8:00?

    (c) What are the x, y components of that 2:00 vector $\boldsymbol{v} = (\cos\theta, \sin\theta)$?

14. **Suppose the twelve vectors start from 6:00 at the bottom instead of $(0,0)$ at the center. The vector to 12:00 is doubled to $(0,2)$. The new twelve vectors add to ____________.**

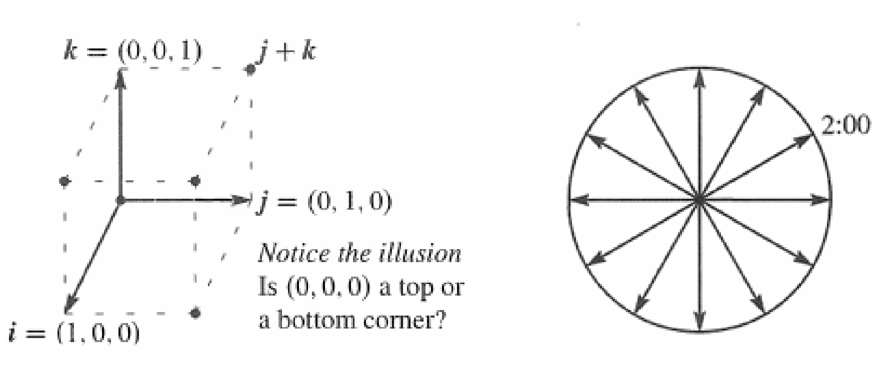


Figure 1.4: Unit cube from i,j, k and twelve clock vectors.

Problems 15–19 go further with linear combinations of $\boldsymbol{v}$ and $\boldsymbol{w}$ (Figure 1.5a).

15. **Figure 1.5a shows $\frac{1}{2}\boldsymbol{v} + \frac{1}{2}\boldsymbol{w}$. Mark the points $\frac{3}{4}\boldsymbol{v} + \frac{1}{4}\boldsymbol{w}$ and $\frac{1}{4}\boldsymbol{v} + \frac{3}{4}\boldsymbol{w}$ and $\boldsymbol{v} + \boldsymbol{w}$.**

16. **Mark the point $-\boldsymbol{v} + 2\boldsymbol{w}$ and any other combination $c\boldsymbol{v} + d\boldsymbol{w}$ with $c + d = 1$. Draw the line of all combinations that have $c + d = 1$.**

17. **Locate $\frac{1}{3}\boldsymbol{v} + \frac{1}{3}\boldsymbol{w}$ and $\frac{2}{3}\boldsymbol{v} + \frac{2}{3}\boldsymbol{w}$. The combinations $c\boldsymbol{v} + c\boldsymbol{w}$ fill out what line?**

18. **Restricted by $0 \le c \le 1$ and $0 \le d \le 1$, shade in all combinations $c\boldsymbol{v} + d\boldsymbol{w}$.**

19. **Restricted only by $c \ge 0$ and $d \ge 0$ draw the "cone" of all combinations $c\boldsymbol{v} + d\boldsymbol{w}$.**

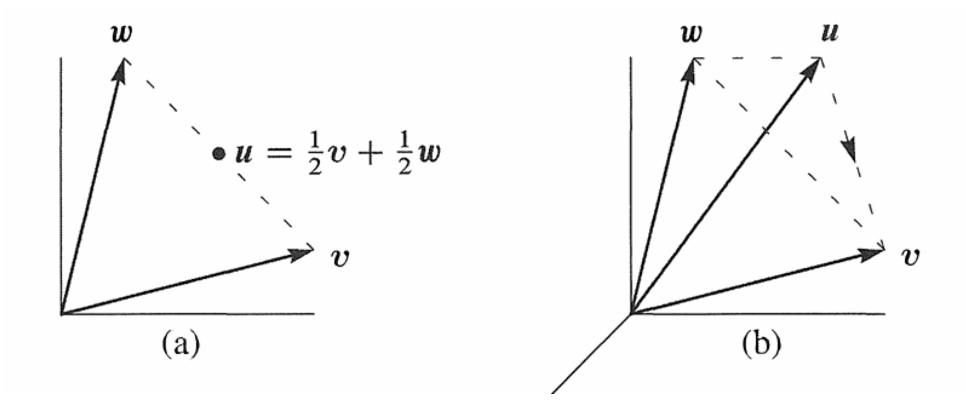


Figure 1.5: Problems 15-19 in a plane   Problems 20-25 in 3-dimensional space

Problems 20-25 deal with $\boldsymbol{u}$, $\boldsymbol{v}$, $\boldsymbol{w}$ in three-dimensional space (see Figure 1.5b).

20. **Locate $\frac{1}{2}\boldsymbol{v} + \frac{1}{2}\boldsymbol{w} + \frac{1}{2}\boldsymbol{v}$ and $\frac{1}{2}\boldsymbol{v} + \frac{1}{2}\boldsymbol{w} + \frac{1}{2}\boldsymbol{w}$ in Figure 1.5b. Challenge problem: Under what restrictions on c, d, e, will the combinations $c\boldsymbol{u} + d\boldsymbol{v} + e\boldsymbol{w}$ fill in the dashed triangle? To stay in the triangle, one requirement is $c \geq 0, d \geq 0, e \geq 0$.**

21. **The three sides of the dashed triangle are $\boldsymbol{v} - \boldsymbol{u}$ and $\boldsymbol{w} - \boldsymbol{v}$ and $\boldsymbol{u} - \boldsymbol{w}$. Their sum is __________. Draw the head-to-tail addition around a plane triangle of $(3, 1)$ plus $(-1, 1)$ plus $(-2, -2)$.**

22. **Shade in the pyramid of combinations $c\boldsymbol{u} + d\boldsymbol{v} + e\boldsymbol{w}$ with $c \geq 0$, $d \geq 0$, $e \geq 0$ and $c + d + e \leq 1$. Mark the vector $\frac{1}{2}(\boldsymbol{u} + \boldsymbol{v} + \boldsymbol{w})$ as inside or outside this pyramid.**

23. **If you look at all combinations of those $\boldsymbol{u}, \boldsymbol{v}$, and $\boldsymbol{w}$, is there any vector that can't be produced from $c\boldsymbol{u} + d\boldsymbol{v} + e\boldsymbol{w}$? Different answer if $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ are all in __________.**

24. **Which vectors are combinations of $\boldsymbol{u}$ and $\boldsymbol{v}$, and also combinations of $\boldsymbol{v}$ and $\boldsymbol{w}$?**

25. **Draw vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ so that their combinations $c\boldsymbol{u} + d\boldsymbol{v} + e\boldsymbol{w}$ fill only a line. Find vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ so that their combinations $c\boldsymbol{u} + d\boldsymbol{v} + e\boldsymbol{w}$ fill only a plane.**

26. **What combination $c\begin{bmatrix}2\\1\end{bmatrix} + d\begin{bmatrix}3\\1\end{bmatrix}$ produces $\begin{bmatrix}14\\8\end{bmatrix}$? Express this question as two equations for the coefficients c and d in the linear combination.**

Challenge Problems

27. **How many corners does a cube have in 4 dimensions? How many 3D faces? How many edges? A typical corner is $(0, 0, 1, 0)$. A typical edge goes to $(0, 1, 0, 0)$.**

28. **Find vectors $\boldsymbol{v}$ and $\boldsymbol{w}$ so that $\boldsymbol{v} + \boldsymbol{w} = (4, 5, 6)$ and $\boldsymbol{v} - \boldsymbol{w} = (2, 5, 8)$. This is a question with __________ unknown numbers, and an equal number of equations to find those numbers.**

29. **Find two different combinations of the three vectors $\boldsymbol{u} = (1, 3)$ and $\boldsymbol{v} = (2, 7)$ and $\boldsymbol{w} = (1, 5)$ that produce $\boldsymbol{b} = (0, 1)$. Slightly delicate question: If I take any three vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ in the plane, will there always be two different combinations that produce $\boldsymbol{b} = (0, 1)$?**

30. **The linear combinations of $\boldsymbol{v} = (a, b)$ and $\boldsymbol{w} = (c, d)$ fill the plane unless __________.**
    Find four vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}, \boldsymbol{z}$ with four components each so that their combinations $c\boldsymbol{u} + d\boldsymbol{v} + e\boldsymbol{w} + f\boldsymbol{z}$ produce all vectors (b_1, b_2, b_3, b_4) in four-dimensional space.

31. **Write down three equations for c, d, e so that $c\boldsymbol{u} + d\boldsymbol{v} + e\boldsymbol{w} = \boldsymbol{b}$. Can you somehow find c, d, e for $\boldsymbol{b}$?**

$$\boldsymbol{u} = \begin{bmatrix}2\\-1\\0\end{bmatrix}, \quad \boldsymbol{v} = \begin{bmatrix}-1\\2\\-1\end{bmatrix}, \quad \boldsymbol{w} = \begin{bmatrix}0\\-1\\2\end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix}1\\0\\0\end{bmatrix}.$$

1.2 Lengths and Dot Products

1. **Calculate the dot products $\boldsymbol{u} \cdot \boldsymbol{v}$ and $\boldsymbol{u} \cdot \boldsymbol{w}$ and $\boldsymbol{u} \cdot (\boldsymbol{v} + \boldsymbol{w})$ and $\boldsymbol{w} \cdot \boldsymbol{v}$:**

$$\boldsymbol{u} = \begin{bmatrix}-.6\\.8\end{bmatrix}, \qquad \boldsymbol{v} = \begin{bmatrix}4\\3\end{bmatrix}, \qquad \boldsymbol{w} = \begin{bmatrix}1\\2\end{bmatrix}.$$

2. **Compute the lengths $\|\boldsymbol{u}\|$ and $\|\boldsymbol{v}\|$ and $\|\boldsymbol{w}\|$ of those vectors. Check the Schwarz inequalities $|\boldsymbol{u} \cdot \boldsymbol{v}| \leq \|\boldsymbol{u}\|\,\|\boldsymbol{v}\|$ and $\boldsymbol{v} \cdot \boldsymbol{w} \leq \|\boldsymbol{v}\|\,\|\boldsymbol{w}\|$.**

3. **Find unit vectors in the directions of $\boldsymbol{v}$ and $\boldsymbol{w}$ in Problem 1, and the cosine of the angle θ. Choose vectors $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$ that make $0°$, $90°$, and $180°$ angles with $\boldsymbol{w}$.**

4. **For any *unit* vectors $\boldsymbol{v}$ and $\boldsymbol{w}$, find the dot products (actual numbers) of**

    (a) $\boldsymbol{v}$ **and** $-\boldsymbol{v}$ **(b)** $\boldsymbol{v} + \boldsymbol{w}$ **and** $\boldsymbol{v} - \boldsymbol{w}$ **(c)** $\boldsymbol{v} - 2\boldsymbol{w}$ **and** $\boldsymbol{v} + 2\boldsymbol{w}$

5. **Find unit vectors $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$ in the directions of $\boldsymbol{v} = (1, 3)$ and $\boldsymbol{w} = (2, 1, 2)$. Find unit vectors $\boldsymbol{U}_1$ and $\boldsymbol{U}_2$ that are perpendicular to $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$.**

6. (a) Describe every vector $w = (w_1, w_2)$ that is perpendicular to $v = (2, -1)$.
   (b) All vectors perpendicular to $V = (1, 1, 1)$ lie on a ____________ in 3 dimensions.
   (c) The vectors perpendicular to both $(1, 1, 1)$ and $(1, 2, 3)$ lie on a ____________.

7. Find the angle θ (from its cosine) between these pairs of vectors:

   (a) $v = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix}$ and $w = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ (b) $v = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$ and $w = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$

   (c) $v = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix}$ and $w = \begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix}$ (d) $v = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$

8. True or false (give a reason if true or find a counterexample if false):

   (a) If $u = (1, 1, 1)$ is perpendicular to v and w, then v is parallel to w.
   (b) If u is perpendicular to v and w, then u is perpendicular to $v + 2w$.
   (c) If u and v are perpendicular unit vectors then $\|u - v\| = \sqrt{2}$. Yes!

9. The slopes of the arrows from $(0, 0)$ to (v_1, v_2) and (w_1, w_2) are v_2/v_1 and w_2/w_1. Suppose the product v_2w_2/v_1w_1 of those slopes is -1. Show that $v \cdot w = 0$ and the vectors are perpendicular. (The line $y = 4x$ is perpendicular to $y = -\frac{1}{4}x$.)

10. Draw arrows from $(0, 0)$ to the points $v = (1, 2)$ and $w = (-2, 1)$. Multiply their slopes. That answer is a signal that $v \cdot w = 0$ and the arrows are ____________.

11. If $v \cdot w$ is negative, what does this say about the angle between v and w? Draw a 3-dimensional vector v (an arrow), and show where to find all w's with $v \cdot w < 0$.

12. With $v = (1, 1)$ and $w = (1, 5)$ choose a number c so that $w - cv$ is perpendicular to v. Then find the formula for c starting from any nonzero v and w.

13. Find nonzero vectors v and w that are perpendicular to $(1, 0, 1)$ and to each other.

14. Find nonzero vectors u, v, w that are perpendicular to $(1, 1, 1)$ and to each other.

15. The geometric mean of $x = 2$ and $y = 8$ is $\sqrt{xy} = 4$. The arithmetic mean is larger: $\frac{1}{2}(x + y) =$ ____________. This would come in Example 6 from the Schwarz inequality for $v = (\sqrt{2}, \sqrt{8})$ and $w = (\sqrt{8}, \sqrt{2})$. Find $\cos\theta$ for this v and w.

16. How long is the vector $v = (1, 1, \ldots, 1)$ in 9 dimensions? Find a unit vector u in the same direction as v and a unit vector w that is perpendicular to v.

17. What are the cosines of the angles α, β, θ between the vector $(1, 0, -1)$ and the unit vectors i, j, k along the axes? Check the formula $\cos^2\alpha + \cos^2\beta + \cos^2\theta = 1$.

Problems 18-28 lead to the main facts about lengths and angles in triangles.

18. The parallelogram with sides $v = (4, 2)$ and $w = (-1, 2)$ is a rectangle. Check the Pythagoras formula $a^2 + b^2 = c^2$ which is for *right triangles only*:

$$\boxed{(\text{length of } v)^2 + (\text{length of } w)^2 = (\text{length of } v + w)^2.}$$

19. (Rules for dot products) These equations are simple but useful:

   (a) $v \cdot w = w \cdot v$ (b) $u \cdot (v + w) = u \cdot v + u \cdot w$ (c) $(cv) \cdot w = c(v \cdot w)$

   Use (2) with $u = v + w$ to prove $\|v + w\|^2 = v \cdot v + 2v \cdot w + w \cdot w$.

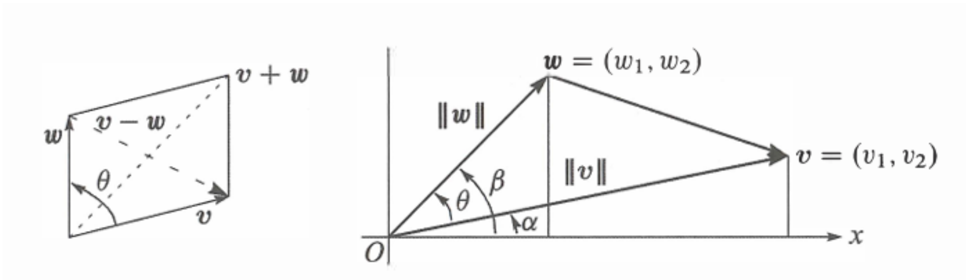
20. The "Law of Cosines" comes from $(v - w) \cdot (v - w) = v \cdot v - 2v \cdot w + w \cdot w$:

$$\text{Cosine Law} \qquad \|v - w\|^2 = \|v\|^2 - 2\|v\|\,\|w\|\cos\theta + \|w\|^2.$$

   Draw a triangle with sides v and w and $v - w$. Which of the angles is θ?

21. **The triangle inequality says: (length of $v + w$) $\leq$ (length of v) + (length of w).**
Problem 19 found $\|v + w\|^2 = \|v\|^2 + 2v \cdot w + \|w\|^2$.
Increase by $\|v\|\,\|w\|$ to show that $v \cdot w$ can not exceed $\|v\|\,\|w\|$:

Triangle inequality $\quad \|v+w\|^2 \leq (\|v\|+\|w\|)^2 \quad$ **or** $\quad \boxed{\|v + w\| \leq \|v\| + \|w\|}$.



22. **The Schwarz inequality $|v \cdot w| \leq \|v\|\,\|w\|$ by algebra instead of trigonometry:**
    (a) **Multiply out both sides of $(v_1w_1 + v_2w_2)^2 \leq (v_1^2 + v_2^2)(w_1^2 + w_2^2)$.**
    (b) **Show that the difference between those two sides equals $(v_1w_2 - v_2w_1)^2$. This cannot be negative since it is a square—so the inequality is true.**

23. **The figure shows that $\cos\alpha = v_1/\|v\|$ and $\sin\alpha = v_2/\|v\|$. Similarly $\cos\beta$ is ______ and $\sin\beta$ is ______. The angle θ is $\beta - \alpha$. Substitute into the trigonometry formula $\cos\beta\cos\alpha + \sin\beta\sin\alpha$ for $\cos(\beta - \alpha)$ to find $\cos\theta = v \cdot w/\|v\|\,\|w\|$.**

24. **One-line proof of the inequality $|u \cdot U| \leq 1$ for unit vectors (u_1, u_2) and (U_1, U_2):**

$$|u \cdot U| \leq |u_1||U_1| + |u_2||U_2| \leq \frac{u_1^2 + U_1^2}{2} + \frac{u_2^2 + U_2^2}{2} = 1.$$

Put $(u_1, u_2) = (.6, .8)$ and $(U_1, U_2) = (.8, .6)$ in that whole line and find $\cos\theta$.

25. **Why is $|\cos\theta|$ never greater than 1 in the first place?**

26. **(Recommended) Draw a parallelogram.**

27. **Parallelogram with two sides v and w. Show that the squared diagonal lengths $\|v+w\|^2+\|v-w\|^2$ add to the sum of four squared side lengths $2\|v\|^2 + 2\|w\|^2$.**

28. **If $v = (1, 2)$ draw all vectors $w = (x, y)$ in the xy-plane with $v \cdot w = x + 2y = 5$. Why do those w's lie along a line? Which is the shortest w?**

29. **(Recommended) If $\|v\| = 5$ and $\|w\| = 3$, what are the smallest and largest possible values of $\|v - w\|$? What are the smallest and largest possible values of $v \cdot w$?**

Challenge Problems

30. **Can three vectors in the xy-plane have $u \cdot v < 0$ and $v \cdot w < 0$ and $u \cdot w < 0$?**
I don't know how many vectors in xyz space can have all negative dot products. (Four of those vectors in the plane would certainly be impossible....)

31. **Pick any numbers that add to $x + y + z = 0$. Find the angle between your vector $v = (x, y, z)$ and the vector $w = (z, x, y)$. Challenge question: Explain why $v \cdot w/\|v\|\,\|w\|$ is always $-\frac{1}{2}$.**

32. **How could you prove $\sqrt[3]{xyz} \leq \frac{1}{3}(x + y + z)$ (geometric mean $\leq$ arithmetic mean)?**

33. **Find 4 perpendicular unit vectors of the form $\left(\pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2}\right)$: Choose $+$ or $-$.**

34. **Using $v = \mathrm{randn}(3, 1)$ in MATLAB, create a random unit vector $u = v/\|v\|$.**
Using $V = \mathrm{randn}(3, 30)$ create 30 more random unit vectors U_j.
What is the average size of the dot products $|u \cdot U_j|$?
In calculus, the average is $\displaystyle\int_0^\pi |\cos\theta|\, d\theta/\pi = 2/\pi$.

1.3 Matrices

1. **Find the linear combination $3s_1 + 4s_2 + 5s_3 = b$. Then write b as a matrix-vector multiplication Sx, with $3, 4, 5$ in x. Compute the three dot products (row of S) $\cdot x$:**

$$s_1 = \begin{bmatrix}1\\1\\1\end{bmatrix}, \quad s_2 = \begin{bmatrix}0\\1\\1\end{bmatrix}, \quad s_3 = \begin{bmatrix}0\\0\\1\end{bmatrix} \quad \text{go into the columns of } S.$$

2. **Solve these equations $Sy = b$ with s_1, s_2, s_3 in the columns of S:**

$$\begin{bmatrix}1&0&0\\1&1&0\\1&1&1\end{bmatrix}\begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix} = \begin{bmatrix}1\\1\\1\end{bmatrix} \quad \text{and} \quad \begin{bmatrix}1&0&0\\1&1&0\\1&1&1\end{bmatrix}\begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix} = \begin{bmatrix}1\\4\\9\end{bmatrix}.$$

S is a sum matrix. The sum of the first 3 odd numbers is ____________.

3. **Solve these three equations for y_1, y_2, y_3 in terms of c_1, c_2, c_3:**

$$Sy = c \qquad \begin{bmatrix}1&0&0\\1&1&0\\1&1&1\end{bmatrix}\begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix} = \begin{bmatrix}c_1\\c_2\\c_3\end{bmatrix}.$$

Write the solution y as a matrix $A = S^{-1}$ times the vector c. Are the columns of S independent or dependent?

4. **Find a combination $x_1w_1 + x_2w_2 + x_3w_3$ that gives the zero vector with $x_1 = 1$:**

$$w_1 = \begin{bmatrix}1\\2\\3\end{bmatrix} \qquad w_2 = \begin{bmatrix}4\\5\\6\end{bmatrix} \qquad w_3 = \begin{bmatrix}7\\8\\9\end{bmatrix}$$

Those vectors are (independent) (dependent). The three vectors lie in a ____________. The matrix W with those three columns is not invertible.

5. **The rows of that matrix W produce three vectors (I write them as columns):**

$$r_1 = \begin{bmatrix}1\\4\\7\end{bmatrix} \qquad r_2 = \begin{bmatrix}2\\5\\8\end{bmatrix} \qquad r_3 = \begin{bmatrix}3\\6\\9\end{bmatrix}$$

Linear algebra says that these vectors must also lie in a plane. There must be many combinations with $y_1r_1 + y_2r_2 + y_3r_3 = 0$. Find two sets of y's.

6. **Which numbers c give dependent columns so a combination of columns equals zero?**

$$\begin{bmatrix}1&1&0\\3&2&1\\7&4&c\end{bmatrix} \quad \begin{bmatrix}1&0&c\\1&1&0\\0&1&1\end{bmatrix} \quad \begin{bmatrix}c&c&c\\2&1&5\\3&3&6\end{bmatrix} \quad \begin{matrix}\textbf{maybe}\\ \textbf{always}\\ \textbf{independent for } c \neq 0\end{matrix}$$

7. **If the columns combine into $Ax = 0$, then each of the rows has $r \cdot x = 0$:**

$$\begin{bmatrix}a_1 & a_2 & a_3\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix} \quad \text{By rows} \quad \begin{bmatrix}r_1\cdot x\\r_2\cdot x\\r_3\cdot x\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}.$$

The three rows also lie in a plane. Why is that plane perpendicular to x?

8. **Moving to a 4×4 difference equation $Ax = b$, find the four components x_1, x_2, x_3, x_4. Then write this solution as $x = A^{-1}b$ to find the inverse matrix:**

$$Ax = \begin{bmatrix}1&0&0&0\\-1&1&0&0\\0&-1&1&0\\0&0&-1&1\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix} = \begin{bmatrix}b_1\\b_2\\b_3\\b_4\end{bmatrix} = b.$$

9. What is the cyclic 4 by 4 difference matrix C? It will have 1 and -1 in each row and each column. Find all solutions $\boldsymbol{x} = (x_1, x_2, x_3, x_4)$ to $C\boldsymbol{x} = \mathbf{0}$. The four columns of C lie in a "three-dimensional hyperplane" inside four-dimensional space.

10. A forward difference matrix Δ is upper triangular:

$$\Delta \boldsymbol{z} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} z_2 - z_1 \\ z_3 - z_2 \\ 0 - z_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \boldsymbol{b}.$$

Find z_1, z_2, z_3 from b_1, b_2, b_3. What is the inverse matrix in $\boldsymbol{z} = \Delta^{-1}\boldsymbol{b}$?

11. Show that the forward differences $(t+1)^2 - t^2$ are $2t + 1 =$ odd numbers. As in calculus, the difference $(t+1)^n - t^n$ will begin with the derivative of t^n, which is __________________.

12. The last lines of the Worked Example say that the 4×4 centered difference matrix in (16) is invertible. Solve $C\boldsymbol{x} = (b_1, b_2, b_3, b_4)$ to find its inverse in $\boldsymbol{x} = C^{-1}\boldsymbol{b}$.

Challenge Problems

13. The very last words say that the 5×5 centered difference matrix is not invertible. Write down the 5 equations $C\boldsymbol{x} = \boldsymbol{b}$. Find a combination of left sides that gives zero. What combination of b_1, b_2, b_3, b_4, b_5 must be zero? (The 5 columns lie on a "4-dimensional hyperplane" in 5-dimensional space. Hard to visualize.)

14. If (a, b) is a multiple of (c, d) with $abcd \neq 0$, show that (a, c) is a multiple of (b, d). This is surprisingly important: two columns are falling on one line. You could use numbers first to see how a, b, c, d are related. The question will lead to:

$$\textbf{If } \begin{bmatrix} a & b \\ c & d \end{bmatrix} \textbf{ has dependent rows, then it also has dependent columns.}$$

2 Chapter 2: Solving Linear Equations

2.1 Vectors and Linear Equations

1. **With $A = I$ (the identity matrix) draw the planes in the row picture. Three sides of a box meet at the solution $\boldsymbol{x} = (x, y, z) = (2, 3, 4)$:**

$$\begin{aligned} 1x + 0y + 0z &= 2 \\ 0x + 1y + 0z &= 3 \\ 0x + 0y + 1z &= 4 \end{aligned} \quad \textbf{or} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}.$$

   Draw the vectors in the column picture. Two times column 1 plus three times column 2 plus four times column 3 equals the right side $\boldsymbol{b}$.

2. **If the equations in Problem 1 are multiplied by $2, 3, 4$ they become $D\boldsymbol{x} = \boldsymbol{B}$:**

$$\begin{aligned} 2x + 0y + 0z &= 4 \\ 0x + 3y + 0z &= 9 \\ 0x + 0y + 4z &= 16 \end{aligned} \quad \textbf{or} \quad D\boldsymbol{x} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \\ 16 \end{bmatrix} = \boldsymbol{B}.$$

   Why is the row picture the same? Is the solution $\boldsymbol{X}$ the same as x? What is changed in the column picture—the columns or the right combination to give $\boldsymbol{B}$?

3. **If equation 1 is added to equation 2, which of these are changed: the planes in the row picture, the vectors in the column picture, the coefficient matrix, the solution? The new equations in Problem 1 would be $x = 2$, $x + y = 5$, $z = 4$.**

4. **Find a point with $z = 2$ on the intersection line of the planes $x + 3z = 6$ and $x - y + z = 4$. Find the point with $z = 0$. Find a third point halfway between.**

5. **The first of these equations plus the second equals the third:**

$$\begin{aligned} x + y + z &= 2 \\ x + 2y + z &= 3 \\ 2x + 3y + 2z &= 5. \end{aligned}$$

   The first two planes meet along a line. The third plane contains that line, because if x, y, z satisfy the first two equations then they also ________________. The equations have infinitely many solutions (the whole line L). Find three solutions on L.

6. **Move the third plane in Problem 5 to a parallel plane $2x + 3y + 2z = 9$. Now the three equations have no solution—why not? The first two planes meet along the line L, but the third plane doesn't ________________ that line.**

7. **In Problem 5 the columns are $(1, 1, 2)$ and $(1, 2, 3)$ and $(1, 1, 2)$. This is a "singular case" because the third column is ________________. Find two combinations of the columns that give $\boldsymbol{b} = (2, 3, 5)$. This is only possible for $\boldsymbol{b} = (4, 6, c)$ if $c =$ __________.**

8. **Normally 4 "planes" in 4-dimensional space meet at a ________________. Normally 4 column vectors in 4-dimensional space can combine to produce $\boldsymbol{b}$. What combination of $(1, 0, 0, 0)$, $(1, 1, 0, 0)$, $(1, 1, 1, 0)$, $(1, 1, 1, 1)$ produces $\boldsymbol{b} = (3, 3, 3, 2)$? What 4 equations for x, y, z, t are you solving?**

Problems 9–14 are about multiplying matrices and vectors.

9. **Compute each Ax by dot products of the rows with the column vector:**

$$\textbf{(a)} \begin{bmatrix} 1 & 2 & 4 \\ -2 & 3 & 1 \\ -4 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} \qquad \textbf{(b)} \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \end{bmatrix}.$$

10. Compute each Ax in Problem 9 as a combination of the columns:

$$9\textbf{(a) becomes} \quad Ax = 2\begin{bmatrix}1\\-2\\-4\end{bmatrix} + 2\begin{bmatrix}2\\3\\1\end{bmatrix} + 3\begin{bmatrix}4\\1\\2\end{bmatrix} = \begin{bmatrix}\;\\\;\\\;\end{bmatrix}.$$

How many separate multiplications for Ax, when the matrix is "3 by 3"?

11. Find the two components of Ax by rows or by columns:

$$\begin{bmatrix}2&3\\5&1\end{bmatrix}\begin{bmatrix}4\\2\end{bmatrix}, \qquad \begin{bmatrix}3&6\\6&12\end{bmatrix}\begin{bmatrix}2\\-1\end{bmatrix}, \qquad \begin{bmatrix}1&2&4\\2&0&1\end{bmatrix}\begin{bmatrix}3\\1\\1\end{bmatrix}.$$

12. Multiply A times x to find three components of Ax:

$$\begin{bmatrix}0&0&1\\0&1&0\\1&0&0\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}, \qquad \begin{bmatrix}2&1&3\\1&2&3\\3&3&6\end{bmatrix}\begin{bmatrix}1\\1\\-1\end{bmatrix}, \qquad \begin{bmatrix}2&1\\1&2\\3&3\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix}.$$

13. (a) A matrix with m rows and n columns multiplies a vector with ____________ components to produce a vector with ____________ components.

(b) The planes from the m equations $Ax = b$ are in ____________-dimensional space. The combination of the columns of A is in ____________-dimensional space.

14. Write $2x + 3y + z + 5t = 8$ as a matrix A (how many rows?) multiplying the column vector $\boldsymbol{x} = (x, y, z, t)$ to produce $\boldsymbol{b}$. The solutions $\boldsymbol{x}$ fill a plane or a "hyperplane" in 4-dimensional space. The plane is 3-dimensional with no 4D volume.

Problems 15–22 ask for matrices that act in special ways on vectors.

15. (a) What is the 2×2 identity matrix? I times $\begin{bmatrix}x\\y\end{bmatrix}$ equals $\begin{bmatrix}x\\y\end{bmatrix}$.

(b) What is the 2×2 exchange matrix? P times $\begin{bmatrix}x\\y\end{bmatrix}$ equals $\begin{bmatrix}y\\x\end{bmatrix}$.

16. (a) What 2×2 matrix R rotates every vector by $90°$? R times $\begin{bmatrix}x\\y\end{bmatrix}$ is $\begin{bmatrix}-y\\x\end{bmatrix}$.

(b) What 2×2 matrix R^2 rotates every vector by $180°$?

17. Find the matrix P that multiplies (x, y, z) to give (y, x, z). Find the matrix Q that multiplies (y, x, z) to bring back (x, y, z).

18. What 2×2 matrix E subtracts the first component from the second component? What 3×3 matrix does the same?

$$E\begin{bmatrix}3\\5\end{bmatrix} = \begin{bmatrix}3\\2\end{bmatrix} \quad \textbf{and} \quad E\begin{bmatrix}3\\5\\7\end{bmatrix} = \begin{bmatrix}3\\2\\7\end{bmatrix}.$$

19. What 3×3 matrix E multiplies (x, y, z) to give $(x, y, z + x)$? What matrix E^{-1} multiplies (x, y, z) to give $(x, y, z - x)$? If you multiply $(3, 4, 5)$ by E and then multiply by E^{-1}, the two results are ____________ and ____________.

20. (a) What 2×2 matrix P_1 projects the vector (x, y) onto the x axis to produce $(x, 0)$?

(b) What matrix P_2 projects onto the y axis to produce $(0, y)$? If you multiply $(5, 7)$ by P_1 and then multiply by P_2, you get ____________ and ____________.

21. What 2×2 matrix R rotates every vector through $45°$? The vector $(1, 0)$ goes to $(\sqrt{2}/2, \sqrt{2}/2)$. The vector $(0, 1)$ goes to $(-\sqrt{2}/2, \sqrt{2}/2)$. Those determine the matrix. Draw these particular vectors in the xy plane and find R.

22. **Write the dot product of $(1,4,5)$ and (x,y,z) as a matrix multiplication Ax. The matrix A has one row. The solutions to $Ax = 0$ lie on a ____________ perpendicular to the vector ____________. The columns of A are only in ____________-dimensional space.**

23. **In MATLAB notation, write the commands that define this matrix A and the column vectors x and b. What command would test whether or not $Ax = b$?**

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad \boldsymbol{x} = \begin{bmatrix} 5 \\ -2 \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}.$$

24. **The MATLAB commands `A = eye(3)` and `v = [3;5]'` produce the 3×3 identity matrix and the column vector $(3,4,5)$. What are the outputs from `A*v` and `v'*A`? (Computer not needed!) If you ask for `v*A`, what happens?**

25. **If you multiply the 4×4 all-ones matrix $A = \texttt{ones}(4)$ and the column vector $\boldsymbol{v} = \texttt{ones}(4,1)$, what is Av? (Computer not needed.) If you multiply $B = \texttt{eye}(4) + \texttt{ones}(4)$ times $\boldsymbol{w} = \texttt{zeros}(4,1) + 2 * \texttt{ones}(4,1)$, what is $B * \boldsymbol{w}$?**

26. **Draw the row and column pictures for the equations $x - 2y = 0$, $x + y = 6$.**

27. **For two linear equations in three unknowns x, y, z, the row picture will show (2 or 3) (lines or planes) in (2 or 3)-dimensional space. The column picture is in (2 or 3)-dimensional space. The solutions normally lie on a __________________.**

28. **For four linear equations in two unknowns x and y, the row picture shows four __________________. The column picture is in __________________-dimensional space. The equations have no solution unless the vector on the right side is a combination of __________________.**

29. **Start with the vector $\boldsymbol{u}_0 = (1,0)$. Multiply again and again by the same "Markov matrix"**

$$A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}.$$

The next three vectors are $\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3$:

$$\boldsymbol{u}_1 = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} .8 \\ .2 \end{bmatrix}, \qquad \boldsymbol{u}_2 = A\boldsymbol{u}_1 = \_\_\_\_\_\_\_\_\_\_\_\_, \qquad \boldsymbol{u}_3 = A\boldsymbol{u}_2 = \_\_\_\_\_\_\_\_\_\_\_\_.$$

What property do you notice for all four vectors $\boldsymbol{u}_0, \boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3$?

Challenge Problems

30. **Continue Problem 29 from $\boldsymbol{u}_0 = (1,0)$ to $\boldsymbol{u}_7$, and also from $\boldsymbol{v}_0 = (0,1)$ to $\boldsymbol{v}_7$. What do you notice about $\boldsymbol{u}_7$ and $\boldsymbol{v}_7$? Here are two MATLAB codes, with `while` and `for`:**

```
u = [1; 0]; A = [.8 .3; .2 .7];
x = u; k = [0 : 7];
while size(x,2) ¡= 7
    u = A*u; x = [x u];
end
plot(k, x)
```

```
v = [0; 1]; A = [.8 .3; .2 .7];
x = v; k = [0 : 7];
for j = 1 : 7
    v = A*v; x = [x v];
end
plot(k, x)
```

The $\boldsymbol{u}$'s and $\boldsymbol{v}$'s are approaching a steady state vector $\boldsymbol{s}$. Guess that vector and check that $A\boldsymbol{s} = \boldsymbol{s}$. If you start with $\boldsymbol{s}$, you stay with $\boldsymbol{s}$.

31. **Invent a 3×3 magic matrix M_3 with entries $1, 2, \ldots, 9$. All rows and columns and diagonals add to 15. The first row could be $8, 3, 4$. What is M_3 times $(1,1,1)$? What is M_4 times $(1,1,1,1)$ if a 4×4 magic matrix has entries $1, \ldots, 16$?**

32. **Suppose $\boldsymbol{u}$ and $\boldsymbol{v}$ are the first two columns of a 3×3 matrix A. Which third columns $\boldsymbol{w}$ would make this matrix singular? Describe a typical column picture of $A\boldsymbol{x} = \boldsymbol{b}$ in that singular case, and a typical row picture (for a random $\boldsymbol{b}$).**

33. **Multiplication by A is a "linear transformation". Those words mean:**

If w is a combination of u and v, then Aw is the same combination of Au and Av.

It is this "linearity" $Aw = cAu + dAv$ that gives us the name "linear algebra".

Problem: If $\boldsymbol{u} = \begin{bmatrix}1\\0\end{bmatrix}$ and $\boldsymbol{v} = \begin{bmatrix}0\\1\end{bmatrix}$, then Au and Av are the columns of A.

Combine $\boldsymbol{w} = c\boldsymbol{u} + d\boldsymbol{v}$. If $\boldsymbol{w} = \begin{bmatrix}5\\7\end{bmatrix}$, how is Aw connected to Au and Av?

34. **Start from the four equations $-x_{i+1} + 2x_i - x_{i-1} = i$ (for $i = 1, 2, 3, 4$ with $x_0 = x_5 = 0$). Write those equations in their matrix form $A\boldsymbol{x} = \boldsymbol{b}$. Can you solve them for x_1, x_2, x_3, x_4?**

35. **A 9×9 Sudoku matrix S has the numbers $1, \ldots, 9$ in every row and every column, and in every 3 by 3 block. For the all-ones vector $\boldsymbol{x} = (1, \ldots, 1)$, what is $S\boldsymbol{x}$?**
A better question is: Which row exchanges will produce another Sudoku matrix? Also, which exchanges of block rows give another Sudoku matrix?
Section 2.7 will look at all possible permutations (reorderings) of the rows. I can see 6 orders for the first 3 rows, all giving Sudoku matrices. Also 6 permutations of the next 3 rows, and of the last 3 rows. And 6 block permutations of the block rows?

2.2 The Idea of Elimination

Problems 1–10 are about elimination on 2×2 systems.

1. **What multiple ℓ_{21} of equation 1 should be subtracted from equation 2?**

$$\begin{aligned} 2x + 3y &= 1 \\ 10x + 9y &= 11 \end{aligned}$$

After elimination, write down the upper triangular system and circle the two pivots. The numbers 1 and 11 don't affect the pivots—use them now in back substitution.

2. **Solve the triangular system of Problem 1 by back substitution, y before x. Verify that x times $(2, 10)$ plus y times $(3, 9)$ equals $(1, 11)$. If the right side changes to $(4, 44)$, what is the new solution?**

3. **What multiple of equation 1 should be subtracted from equation 2?**

$$\begin{aligned} 2x - 4y &= 6 \\ -x + 5y &= 0 \end{aligned}$$

After this elimination step, solve the triangular system. If the right side changes to $(-6, 0)$, what is the new solution?

4. **What multiple ℓ of equation 1 should be subtracted from equation 2 to remove c?**

$$\begin{aligned} ax + by &= f \\ cx + dy &= g \end{aligned}$$

The first pivot is a (assumed nonzero). Elimination produces what formula for the second pivot? What is y? The second pivot is missing when $ad = bc$: singular.

5. **Choose a right side which gives no solution and another right side which gives infinitely many solutions. What are two of those solutions?**

Singular system

$$\begin{aligned} 3x + 2y &= 10 \\ 6x + 4y &= \end{aligned}$$

6. **Choose a coefficient b that makes this system singular. Then choose a right side g that makes it solvable. Find two solutions in that singular case.**

$$\begin{aligned} 2x + by &= 16 \\ 4x + 8y &= g \end{aligned}$$

7. **For which numbers a does elimination break down (1) permanently (2) temporarily?**

$$\begin{aligned} ax + 3y &= -3 \\ 4x + 6y &= 6 \end{aligned}$$

Solve for x and y after fixing the temporary breakdown by a row exchange.

8. **For which three numbers k does elimination break down? Which is fixed by a row exchange? In each case, is the number of solutions 0 or 1 or ∞?**

$$\begin{aligned} kx + 3y &= 6 \\ 3x + ky &= -6 \end{aligned}$$

9. **What test on b_1 and b_2 decides whether these two equations allow a solution? How many solutions will they have? Draw the column picture for $\boldsymbol{b} = (1, 2)$ and $(1, 0)$.**

$$\begin{aligned} 3x - 2y &= b_1 \\ 6x - 4y &= b_2 \end{aligned}$$

10. **In the xy plane, draw the lines $x+y = 5$ and $x+2y = 6$ and the equation $y =$ ____ that comes from elimination. The line $5x - 4y = c$ will go through the solution of these equations if $c =$ ____.**

Problems 11–20 study elimination on 3×3 systems (and possible failure).

11. **(Recommended) A system of linear equations can't have exactly two solutions. Why?**
(a) If (x, y, z) and (X, Y, Z) are two solutions, what is another solution?
(b) If 25 planes meet at two points, where else do they meet?

12. **Reduce this system to upper triangular form by two row operations:**

$$\begin{aligned} 2x + 3y + z &= 8 \\ 4x + 7y + 5z &= 20 \\ -2y + 2z &= 0 \end{aligned}$$

Circle the pivots. Solve by back substitution for z, y, x.

13. **Apply elimination (circle the pivots) and back substitution to solve**

$$\begin{aligned} 2x - 3y \phantom{{}+z} &= 3 \\ 4x - 5y + z &= 7 \\ 2x - y - 3z &= 5 \end{aligned}$$

List the three row operations: Subtract ____ times row ____ from row ____.

14. **Which number d forces a row exchange, and what is the triangular system (not singular) for that d? Which d makes this system singular (no third pivot)?**

$$\begin{aligned} 2x + 5y + z &= 0 \\ 4x + dy + z &= 2 \\ y - z &= 3 \end{aligned}$$

15. **Which number b leads later to a row exchange? Which b leads to a missing pivot? In that singular case find a nonzero solution x, y, z.**

$$\begin{aligned} x + by \phantom{{}-z} &= 0 \\ x - 2y - z &= 0 \\ +\, y + z &= 0 \end{aligned}$$

16. **(a) Construct a 3×3 system that needs two row exchanges to reach a triangular form and a solution.**

    (b) Construct a 3×3 system that needs a row exchange to keep going, but breaks down later.

17. **If rows 1 and 2 are the same, how far can you get with elimination (allowing row exchange)? If columns 1 and 2 are the same, which pivot is missing?**

$$\begin{matrix} \textit{Equal} \\ \textit{rows} \end{matrix} \quad \begin{aligned} 2x - y + z &= 0 \\ 2x - y + z &= 0 \\ 4x + y + z &= 2 \end{aligned} \qquad \begin{aligned} 2x + 2y + z &= 0 \\ 4x + 4y + z &= 0 \\ 6x + 6y + z &= 2 \end{aligned} \quad \begin{matrix} \textit{Equal} \\ \textit{columns} \end{matrix}$$

18. **Construct a 3×3 example that has 9 different coefficients on the left side, but rows 2 and 3 become zero in elimination. How many solutions to your system with $\boldsymbol{b} = (1, 10, 100)$ and how many with $\boldsymbol{b} = (0, 0, 0)$?**

19. **Which number q makes this system singular and which right side t gives it infinitely many solutions? Find the solution that has $z = 1$.**

$$\begin{aligned} x + 4y - 2z &= 1 \\ x + 7y - 6z &= 6 \\ 3y + qz &= t \end{aligned}$$

20. **Three planes can fail to have an intersection point, even if no planes are parallel. The system is singular if row 3 of A is a ______ of the first two rows. Find a third equation that can't be solved together with $x + y + z = 0$ and $x - 2y - z = 1$.**

21. **Find the pivots and the solution for both systems ($A\boldsymbol{x} = \boldsymbol{b}$ and $K\boldsymbol{x} = \boldsymbol{b}$):**

$$\begin{aligned} 2x + y \qquad\quad &= 0 \\ x + 2y + z \qquad &= 0 \\ y + 2z + t &= 0 \\ z + 2t &= 5 \end{aligned} \qquad \begin{aligned} 2x - y \qquad\quad &= 0 \\ -x + 2y - z \qquad &= 0 \\ -y + 2z - t &= 0 \\ -z + 2t &= 5 \end{aligned}$$

22. **If you extend Problem 21 following the $1, 2, 1$ pattern or the $-1, 2, -1$ pattern, what is the fifth pivot? What is the nth pivot? K is my favorite matrix.**

23. **If elimination leads to $x + y = 1$ and $2y = 3$, find three possible original problems.**

24. **For which two numbers a will elimination fail on $A = \begin{bmatrix} a & 2 \\ a & a \end{bmatrix}$?**

25. **For which three numbers a will elimination fail to give three pivots?**

$$A = \begin{bmatrix} a & 2 & 3 \\ a & a & 4 \\ a & a & a \end{bmatrix} \ \textbf{is singular for three values of } a.$$

26. **Look for a matrix that has row sums 4 and 8, and column sums 2 and s:**

$$\textbf{Matrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \qquad \begin{cases} a + b = 4 \\ a + c = 2 \\ c + d = 8 \\ b + d = s \end{cases}$$

The four equations are solvable only if $s =$ ____. Then find two different matrices that have the correct row and column sums. *Extra credit:* Write down the 4×4 system $A\boldsymbol{x} = \boldsymbol{b}$ with $\boldsymbol{x} = (a, b, c, d)$ and make A triangular by elimination.

27. **Elimination in the usual order gives what matrix U and what solution to this "lower triangular" system? We are really solving by *forward substitution*:**

$$\begin{aligned} 3x \quad\quad\quad &= 3 \\ 6x + 2y \quad\quad &= 8 \\ 9x - 2y + z &= 9 \end{aligned}$$

28. **Create a MATLAB command `A(2,:) = ...` for the new row 2, to subtract 3 times row 1 from the existing row 2 if the matrix A is already known.**

Challenge Problems

29. **Find experimentally the average 1st and 2nd and 3rd pivot sizes from MATLAB's `[L,U] = lu(rand(3))`. The average size `abs(U(1,1))` is above $\frac{1}{2}$ because `lu` picks the largest available pivot in column 1. Here $A =$ `rand(3)` has random entries between 0 and 1.**

30. **If the last corner entry is $A(5,5) = 11$ and the last pivot of A is $U(5,5) = 4$, what different entry $A(5,5)$ would have made A singular?**

31. **Suppose elimination takes A to U without row exchanges. Then row j of U is a combination of which rows of A? If $Ax = 0$, is $Ux = 0$? If $Ax = b$, is $Ux = b$? If A starts out lower triangular, what is the upper triangular U?**

32. **Start with 100 equations $Ax = 0$ for 100 unknowns $x = (x_1, \ldots, x_{100})$. Suppose elimination reduces the 100th equation to $0 = 0$, so the system is "singular".**

    (a) **Elimination takes linear combinations of the rows. So this singular system has the singular property: Some linear combination of the 100 rows is ______.**

    (b) **Singular systems $Ax = 0$ have infinitely many solutions. This means that some linear combination of the 100 ____ is ______.**

    (c) **Invent a 100 by 100 singular matrix with no zero entries.**

    (d) **For your matrix, describe in words the row picture and the column picture of $Ax = 0$. Not necessary to draw 100-dimensional space.**

2.3 Elimination Using Matrices

Problems 1–15 are about elimination matrices.

1. **Write down the 3 by 3 matrices that produce these elimination steps:**

    (a) **E_{21} subtracts 5 times row 1 from row 2.**

    (b) **E_{32} subtracts -7 times row 2 from row 3.**

    (c) **P exchanges rows 1 and 2, then rows 2 and 3.**

2. **In Problem 1, applying E_{21} and then E_{32} to $\boldsymbol{b} = (1,0,0)$ gives $E_{32}E_{21}\boldsymbol{b} =$ ______. Applying E_{32} before E_{21} gives $E_{21}E_{32}\boldsymbol{b} =$ ______. When E_{32} comes first, row ____ feels no effect from row ____.**

3. **Which three matrices E_{21}, E_{31}, E_{32} put A into triangular form U?**

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 4 & 6 & 1 \\ -2 & 2 & 0 \end{bmatrix} \quad \textbf{and} \quad E_{32}E_{31}E_{21}A = U.$$

    Multiply those E's to get one matrix M that does elimination: $MA = U$.

4. **Include $\boldsymbol{b} = (1,0,0)$ as a fourth column in Problem 3 to produce $[A \mid \boldsymbol{b}]$. Carry out the elimination steps on this augmented matrix to solve $Ax = \boldsymbol{b}$.**

5. **Suppose $a_{33} = 7$ and the third pivot is 5. If you change a_{33} to 11, the third pivot is ______. If you change a_{33} to ______, there is no third pivot.**

6. **If every column of A is a multiple of $(1,1,1)$, then Ax is always a multiple of $(1,1,1)$. Do a 3 by 3 example. How many pivots are produced by elimination?**

7. **Suppose E subtracts 7 times row 1 from row 3.**
   (a) **To invert that step you should ____ 7 times row ____ to row ____.**
   (b) **What "inverse matrix" E^{-1} takes that reverse step (so $E^{-1}E = I$)?**
   (c) **If the reverse step is applied first (and then E) show that $EE^{-1} = I$.**

8. **The determinant of $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is $\det M = ad - bc$. Subtract ℓ times row 1 from row 2 to produce a new M^*. Show that $\det M^* = \det M$ for every ℓ. When $\ell = c/a$, the product of pivots equals the determinant: $(a)(d - \ell b)$ equals $ad - bc$.**

9. (a) **E_{21} subtracts row 1 from row 2 and then P_{23} exchanges rows 2 and 3. What matrix $M = P_{23}E_{21}$ does both steps at once?**
   (b) **P_{23} exchanges rows 2 and 3 and then E_{31} subtracts row 1 from row 3. What matrix $M = E_{31}P_{23}$ does both steps at once? Explain why the M's are the same but the E's are different.**

10. (a) **What 3 by 3 matrix E_{13} will add row 3 to row 1?**
    (b) **What matrix adds row 1 to row 3 and *at the same time* row 3 to row 1?**
    (c) **What matrix adds row 1 to row 3 and *then* adds row 3 to row 1?**

11. **Create a matrix that has $a_{11} = a_{22} = a_{33} = 1$ but elimination produces two negative pivots without row exchanges. (The first pivot is 1.)**

12. **Multiply these matrices:**

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad \textbf{and} \quad \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 1 \\ 1 & 4 & 0 \end{bmatrix}.$$

13. **Explain these facts. If the third column of B is all zero, the third column of EB is all zero (for any E). If the third row of B is all zero, the third row of EB might *not* be zero.**

14. **This 4 by 4 matrix will need elimination matrices E_{21} and E_{32} and E_{43}. What are those matrices?**

$$A = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}.$$

15. **Write down the 3 by 3 matrix that has $a_{ij} = 2i - 3j$. This matrix has $a_{32} = 0$, but elimination still needs E_{32} to produce a zero in the $(3,2)$ position. Which previous step destroys the original zero and what is E_{32}?**

Problems 16–23 are about creating and multiplying matrices.

16. **Write these ancient problems in a 2 by 2 matrix form $Ax = \boldsymbol{b}$ and solve them:**
    (a) **X is twice as old as Y and their ages add to 33.**
    (b) **$(x,y) = (2,5)$ and $(3,7)$ lie on the line $y = mx + c$. Find m and c.**

17. **The parabola $y = a + bx + cx^2$ goes through the points $(x,y) = (1,4)$ and $(2,8)$ and $(3,14)$. Find and solve a matrix equation for the unknowns (a,b,c).**

18. **Multiply these matrices in the orders EF and FE:**

$$E = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & 0 & 1 \end{bmatrix}, \qquad F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & c & 1 \end{bmatrix}.$$

    Also compute $E^2 = EE$ and $F^3 = FFF$. You can guess P^{100}.

19. **Multiply these row exchange matrices in the orders PQ and QP and P^2:**

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \textbf{and} \quad Q = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

Find another non-diagonal matrix whose square is $M^2 = I$.

20. **(a) Suppose all columns of B are the same. Then all columns of EB are the same, because each one is E times ______.**

    (b) Suppose all rows of B are $[\,1\ \ 2\ \ 4\,]$. Show by example that all rows of EB are *not* $[\,1\ \ 2\ \ 4\,]$. It is true that those rows are ________.

21. **If E adds row 1 to row 2 and F adds row 2 to row 1, does EF equal FE?**

22. **The entries of A and $\boldsymbol{x}$ are a_{ij} and x_j. So the first component of $A\boldsymbol{x}$ is $\sum a_{1j}x_j = a_{11}x_1 + \cdots + a_{1n}x_n$. If E_{21} subtracts row 1 from row 2, write a formula for**

    (a) the third component of $E_{21}A\boldsymbol{x}$

    (b) the $(2,1)$ entry of $E_{21}A$

    (c) the $(2,1)$ entry of $E_{21}(E_{21}A)$

    (d) the first component of $(E_{21}E_{21})A\boldsymbol{x}$

23. **The elimination matrix $E = \begin{bmatrix} -2 & 0 \\ 0 & 1 \end{bmatrix}$ subtracts 2 times row 1 of A from row 2 of A. The result is EA. What is the effect of $E(AE)$? In the opposite order AE, we are subtracting 2 times ____ of A from ____. (Do examples.)**

Problems 24–27 include the column $\boldsymbol{b}$ in the augmented matrix $[\,A \mid \boldsymbol{b}\,]$.

24. **Apply elimination to the 2 by 3 augmented matrix $[\,A \mid \boldsymbol{b}\,]$. What is the triangular system $U\boldsymbol{x} = \boldsymbol{c}$? What is the solution $\boldsymbol{x}$?**

$$A\boldsymbol{x} = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 17 \end{bmatrix}.$$

25. **Apply elimination to the 3 by 4 augmented matrix $[\,A \mid \boldsymbol{b}\,]$. How do you know this system has no solution? Change the last number 6 so there is a solution.**

$$A\boldsymbol{x} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix}.$$

26. **The equations $A\boldsymbol{x} = \boldsymbol{b}$ and $A\boldsymbol{x}^* = \boldsymbol{b}^*$ have the same matrix A. What double augmented matrix should you use in elimination to solve both equations at once? Solve both of these equations by working on a 2 by 4 matrix:**

$$\begin{bmatrix} 1 & 4 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \textbf{and} \quad \begin{bmatrix} 1 & 4 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

27. **Choose the numbers a, b, c, d in this augmented matrix so that there is (a) no solution (b) infinitely many solutions.**

$$[A \mid \boldsymbol{b}] = \begin{bmatrix} 1 & 2 & 3 & a \\ 0 & 4 & 5 & b \\ 0 & 0 & d & c \end{bmatrix}.$$

Which of the numbers $a, b, c,$ or d have no effect on the solvability?

28. **If $AB = I$ and $BC = I$ use the associative law to prove $A = C$.**

Challenge Problems

29. Find the triangular matrix E that reduces "Pascal's matrix" to a smaller Pascal:

$$\textbf{Elimination on column 1} \quad E\begin{bmatrix}1&0&0&0\\1&1&0&0\\1&2&1&0\\1&3&3&1\end{bmatrix}=\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&1&1&0\\0&1&2&1\end{bmatrix}.$$

Which matrix M (multiplying several E's) reduces Pascal all the way to I? Pascal's triangular matrix is exceptional, all of its multipliers are $\ell_{ij}=1$.

30. Write $M=\begin{bmatrix}3&4\\6&7\end{bmatrix}$ as a product of many factors $A=\begin{bmatrix}1&0\\1&1\end{bmatrix}$ and $B=\begin{bmatrix}1&1\\0&1\end{bmatrix}$.

(a) What matrix E subtracts row 1 from row 2 to make row 2 of EM smaller?

(b) What matrix F subtracts row 2 of EM from row 1 to reduce row 1 of FEM?

(c) Continue E's and F's until (many E's and F's times) M is (A or B).

(d) E and F are the inverses of A and B! Moving all E's and F's to the right side will give you the desired result M = product of A's and B's. This is possible for integer matrices $M=\begin{bmatrix}a&b\\c&d\end{bmatrix}>0$ that have $ad-bc=1$.

31. Find elimination matrices E_{21} then E_{32} then E_{43} to change K into U:

$$E_{43}\,E_{32}\,E_{21}\begin{bmatrix}1&0&0&0\\-a&1&0&0\\0&-b&1&0\\0&0&-c&1\end{bmatrix}=I.$$

Apply those three steps to the identity matrix I, to multiply $E_{43}E_{32}E_{21}$.

2.4 Rules for Matrix Operations

Problems 1–16 are about the laws of matrix multiplication.

1. A is 3 by 5, B is 5 by 3, C is 5 by 1, and D is 3 by 1. All entries are 1. Which of these matrix operations are allowed, and what are the results?

$$BA \qquad AB \qquad ABD \qquad DC \qquad A(B+C).$$

2. What rows or columns or matrices do you multiply to find

(a) the second column of AB?

(b) the first row of AB?

(c) the entry in row 3, column 5 of AB?

(d) the entry in row 1, column 1 of CDE?

3. Add AB to AC and compare with $A(B+C)$:

$$A=\begin{bmatrix}1&5\\2&3\end{bmatrix} \quad \textbf{and} \quad B=\begin{bmatrix}0&2\\0&1\end{bmatrix} \quad \textbf{and} \quad C=\begin{bmatrix}3&1\\0&0\end{bmatrix}.$$

4. In Problem 3, multiply A times BC. Then multiply AB times C.

5. Compute A^2 and A^3. Make a prediction for A^5 and A^n:

$$A=\begin{bmatrix}1&b\\0&1\end{bmatrix} \quad \textbf{and} \quad A=\begin{bmatrix}2&2\\0&0\end{bmatrix}.$$

6. **Show that $(A+B)^2$ is different from $A^2+2AB+B^2$, when**

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \quad \textbf{and} \quad B = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}.$$

Write down the correct rule for $(A+B)(A+B) = A^2 + \underline{\qquad} + B^2$.

7. **True or false. Give a specific example when false:**

   (a) If columns 1 and 3 of B are the same, so are columns 1 and 3 of AB.

   (b) If rows 1 and 3 of B are the same, so are rows 1 and 3 of AB.

   (c) If rows 1 and 3 of A are the same, so are rows 1 and 3 of ABC.

   (d) $(AB)^2 = A^2B^2$.

8. **How is each row of DA and EA related to the rows of A, when**

$$D = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix} \quad \textbf{and} \quad E = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \quad \textbf{and} \quad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}?$$

How is each column of AD and AE related to the columns of A?

9. **Row 1 of A is added to row 2. This gives EA below. Then column 1 of EA is added to column 2 to produce $(EA)F$:**

$$EA = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ a+c & b+d \end{bmatrix}, \quad (EA)F = (EA) \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & a+b \\ a+c & a+c+b+d \end{bmatrix}.$$

   (a) Do those steps in the opposite order. First add column 1 of A to column 2 by AF, then add row 1 of AF to row 2 by $E(AF)$.

   (b) Compare with $(EA)F$. What law is obeyed by matrix multiplication?

10. **Row 1 of A is again added to row 2 to produce EA. Then F adds row 2 of EA to row 1. The result is $F(EA)$:**

$$F(EA) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ a+c & b+d \end{bmatrix} = \begin{bmatrix} 2a+c & 2b+d \\ a+c & b+d \end{bmatrix}.$$

   (a) Do those steps in the opposite order: first add row 2 to row 1 by FA, then add row 1 of FA to row 2.

   (b) What law is or is not obeyed by matrix multiplication?

11. **This fact still amazes me. If you do a row operation on A and then a column operation, the result is the same as if you did the column operation first. (Try it.) Why is this true?**

12. **(3 by 3 matrices) Choose the only B so that for every matrix A**

   (a) $BA = 4A$

   (b) $BA = 4B$

   (c) BA has rows 1 and 3 of A reversed and row 2 unchanged

   (d) All rows of BA are the same as row 1 of A.

13. **Suppose $AB = BA$ and $AC = CA$ for these two particular matrices B and C:**

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \textbf{commutes with} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \textbf{and} \quad C = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Prove that $a = d$ and $b = c = 0$. Then A is a multiple of I. The only matrices that commute with B and C and all other 2 by 2 matrices are A = multiple of I.

14. **Which of the following matrices are guaranteed to equal $(A-B)^2$: $A^2 - B^2$, $(B-A)^2$, $A^2 - 2AB + B^2$, $A(A-B) - B(A-B)$, $A^2 - AB - BA + B^2$?**

15. **True or false:**

(a) If A^2 is defined then A is necessarily square.

(b) If AB and BA are defined then A and B are square.

(c) If AB and BA are defined then AB and BA are square.

(d) If $AB = B$ then $A = I$.

16. If A is m by n, how many separate multiplications are involved when

(a) A multiplies a vector x with n components?

(b) A multiplies an n by p matrix B?

(c) A multiplies itself to produce A^2? Here $m = n$.

17. For $A = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 6 & 0 \end{bmatrix}$, compute these answers *and nothing more*:

(a) column 2 of AB

(b) row 2 of AB

(c) row 2 of $AA = A^2$

(d) row 2 of $AAA = A^3$.

Problems 18–20 use a_{ij} for the entry in row i, column j of A.

18. Write down the 3 by 3 matrix A whose entries are

(a) $a_{ij} = \min$ **of** i **and** j

(b) $a_{ij} = (-1)^{i+j}$

(c) $a_{ij} = i/j$.

19. What words would you use to describe each of these classes of matrices? Give a 3 by 3 example in each class. Which matrix belongs to all four classes?

(a) $a_{ij} = 0$ **if** $i \neq j$

(b) $a_{ij} = 0$ **if** $i < j$

(c) $a_{ij} = a_{ji}$

(d) $a_{ij} = a_{1j}$.

20. The entries of A are a_{ij}. Assuming that zeros don't appear, what is

(a) the first pivot?

(b) the multiplier ℓ_{31} of row 1 to be subtracted from row 3?

(c) the new entry that replaces a_{32} after that subtraction?

(d) the second pivot?

Problems 21–24 involve powers of A.

21. Compute A^2, A^3, A^4 and also Av, A^2v, A^3v, A^4v for

$$A = \begin{bmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \textbf{and} \quad \boldsymbol{v} = \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix}.$$

22. By trial and error find real nonzero 2 by 2 matrices such that

$$A^2 = -I \qquad BC = 0 \qquad DE = -ED \quad \textbf{(not allowing } DE = 0\textbf{)}.$$

23. (a) Find a nonzero matrix A for which $A^2 = 0$.

(b) Find a matrix that has $A^2 \neq 0$ but $A^3 = 0$.

24. By experiment with $n = 2$ and $n = 3$ predict A^n for these matrices:

$$A_1 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \quad \textbf{and} \quad A_2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad \textbf{and} \quad A_3 = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}.$$

Problems 25–31 use column-row multiplication and block multiplication.

25. Multiply A times I using columns of A (3 by 3) times rows of I.

26. Multiply AB using columns times rows:

$$AB = \begin{bmatrix} 1 & 0 \\ 2 & 4 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 3 & 0 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \begin{bmatrix} 3 & 3 & 0 \end{bmatrix} + \underline{\qquad} = \underline{\qquad}.$$

27. Show that the product of upper triangular matrices is always upper triangular:

$$AB = \begin{bmatrix} x & x & x \\ 0 & x & x \\ 0 & 0 & x \end{bmatrix} \begin{bmatrix} x & x & x \\ 0 & x & x \\ 0 & 0 & x \end{bmatrix} = \begin{bmatrix} \underline{\quad} & \underline{\quad} & \underline{\quad} \\ 0 & \underline{\quad} & \underline{\quad} \\ 0 & 0 & \underline{\quad} \end{bmatrix}.$$

Proof using dot products (Row times column) (Row 2 of A) $\cdot$ (column 1 of B) $= 0$. Which other dot products give zeros?
Proof using full matrices (Column times row) Draw x's and 0's in (column 2 of A) times (row 2 of B). Also show (column 3 of A) times (row 3 of B).

28. Draw the cuts in A (2 by 3) and B (3 by 4) and AB to show how each of the four multiplication rules is really a block multiplication:

(1) Matrix A times columns of B	**Columns of AB**
(2) Rows of A times the matrix B	**Rows of AB**
(3) Rows of A times columns of B	**Inner products (numbers in AB)**
(4) Columns of A times rows of B	**Outer products (matrices add to AB)**

29. Which matrices E_{21} and E_{31} produce zeros in the $(2, 1)$ and $(3, 1)$ positions of $E_{21}A$ and $E_{31}A$?

$$A = \begin{bmatrix} 2 & 1 & 0 \\ -2 & 0 & 1 \\ 8 & 5 & 3 \end{bmatrix}.$$

Find the single matrix $E = E_{31}E_{21}$ that produces both zeros at once. Multiply EA.

30. Block multiplication says that column 1 is eliminated by

$$EA = \begin{bmatrix} 1 & 0 \\ -c/a & I \end{bmatrix} \begin{bmatrix} a & b \\ c & D \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & D - cb/a \end{bmatrix}.$$

In Problem 29, what numbers go into c and D and what is $D - cb/a$?

31. With $i^2 = -1$, the product of $(A + iB)$ and $(x + iy)$ is $Ax + iBx + iAy - By$. Use blocks to separate the real part without i from the imaginary part that multiplies i:

$$\begin{bmatrix} A & -B \\ ? & ? \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} Ax - By \\ ? \end{bmatrix} \quad \begin{matrix} \textbf{real part} \\ \textbf{imaginary part} \end{matrix}$$

(Fill in the ?'s.)

32. (Very important) Suppose you solve $A\boldsymbol{x} = \boldsymbol{b}$ for three special right sides $\boldsymbol{b}$:

$$A\boldsymbol{x}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \textbf{and} \quad A\boldsymbol{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \textbf{and} \quad A\boldsymbol{x}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

If the three solutions $\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3$ are the columns of a matrix X, what is A times X?

33. If the three solutions in Question 32 are $\boldsymbol{x}_1 = (1, 1, 1)$ and $\boldsymbol{x}_2 = (0, 1, 1)$ and $\boldsymbol{x}_3 = (0, 0, 1)$, solve $A\boldsymbol{x} = \boldsymbol{b}$ when $\boldsymbol{b} = (3, 5, 8)$. Challenge problem: What is A?

34. Find all matrices $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ that satisfy $A \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} A$.

35. Suppose a "circle graph" has 4 nodes connected (in both directions) by edges around a circle. What is its adjacency matrix S from Worked Example 2.4 A? What is S^2? Find all the 2-step paths predicted by S^2.

Challenge Problems

36. Practical question Suppose A is m by n, B is n by p, and C is p by q. Then the multiplication count is mnp for $AB + mpq$ for $(AB)C$. The same matrix comes from A times BC with $mnq + npq$ separate multiplications. Notice npq for BC.

(a) If A is 2 by 4, B is 4 by 7, and C is 7 by 10, do you prefer $(AB)C$ or $A(BC)$?

(b) With N-component vectors, would you choose $(\boldsymbol{u}^T\boldsymbol{v})\boldsymbol{w}^T$ or $\boldsymbol{u}^T(\boldsymbol{v}\boldsymbol{w}^T)$?

(c) Divide by mnp to show that $(AB)C$ is faster when $n^{-1} + q^{-1} < m^{-1} + p^{-1}$.

37. To prove that $(AB)C = A(BC)$, use the column vectors $\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n$ of B. First suppose that C has only one column $\boldsymbol{c}_1, \ldots, \boldsymbol{c}_n$.
AB has columns $A\boldsymbol{b}_1, \ldots, A\boldsymbol{b}_n$ and then $(AB)C$ equals $c_1 A\boldsymbol{b}_1 + \cdots + c_n A\boldsymbol{b}_n$.
BC has one column $c_1\boldsymbol{b}_1 + \cdots + c_n\boldsymbol{b}_n$ and then $A(BC)$ equals $A(c_1\boldsymbol{b}_1 + \cdots + c_n\boldsymbol{b}_n)$.
Linearity gives equality of those two sums. This proves $(AB)C = A(BC)$. The same is true for all other ______ of C. Therefore $(AB)C = A(BC)$. Apply to inverses: If $BA = I$ and $AC = I$, prove that the left-inverse B equals the right-inverse C.

38. (a) Suppose A has rows $\boldsymbol{a}_1^T, \ldots, \boldsymbol{a}_m^T$. Why does A^TA equal $\boldsymbol{a}_1\boldsymbol{a}_1^T + \cdots + \boldsymbol{a}_m\boldsymbol{a}_m^T$?

(b) If C is a diagonal matrix with $c_1, \ldots, c_m$ on its diagonal, find a similar sum of columns times rows for A^TCA. First do an example with $m = n = 2$.

2.5 Inverse Matrices

1. Find the inverses (directly or from the 2 by 2 formula) of A, B, C:

$$A = \begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & 0 \\ 4 & 2 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 3 & 4 \\ 5 & 7 \end{bmatrix}.$$

2. For these "permutation matrices" find P^{-1} by trial and error (with 1's and 0's):

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

3. Solve for the first column (x, y) and second column (t, z) of A^{-1}:

$$\begin{bmatrix} 10 & 20 \\ 20 & 50 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 10 & 20 \\ 20 & 50 \end{bmatrix} \begin{bmatrix} t \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

4. Show that $\begin{bmatrix} 3 & 2 \\ 1 & 6 \end{bmatrix}$ is not invertible by trying to solve $AA^{-1} = I$ for column 1 of A^{-1}:

$$\begin{bmatrix} 3 & 2 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \left(\textbf{For a different } A, \textbf{ could column 1 of } A^{-1} \textbf{ be possible to find but not column 2?}\right)$$

5. Find an upper triangular U (not diagonal) with $U^2 = I$ which gives $U = U^{-1}$.

6. (a) If A is invertible and $AB = AC$, prove quickly that $B = C$.

(b) **If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, find two different matrices such that $AB = AC$.**

7. **(Important) If A has row 1 + row 2 = row 3, show that A is not invertible:**
   (a) **Explain why $Ax = (0, 0, 1)$ cannot have a solution. Add eqn 1 + eqn 2.**
   (b) **Which right sides (b_1, b_2, b_3) might allow a solution to $Ax = b$?**
   (c) **In elimination, what happens to equation 3?**

8. **If A has column 1 + column 2 = column 3, show that A is not invertible:**
   (a) **Find a nonzero solution x to $Ax = \mathbf{0}$. The matrix is 3 by 3.**
   (b) **Elimination keeps column 1 + column 2 = column 3. Explain why there is no third pivot.**

9. **Suppose A is invertible and you exchange its first two rows to reach B. Is the new matrix B invertible? How would you find B^{-1} from A^{-1}?**

10. **Find the inverses (in any legal way) of**

$$A = \begin{bmatrix} 0 & 0 & 0 & 2 \\ 0 & 0 & 3 & 0 \\ 0 & 4 & 0 & 0 \\ 5 & 0 & 0 & 0 \end{bmatrix} \quad \textbf{and} \quad B = \begin{bmatrix} 3 & 2 & 0 & 0 \\ 4 & 3 & 0 & 0 \\ 0 & 0 & 6 & 5 \\ 0 & 0 & 7 & 6 \end{bmatrix}.$$

11. (a) **Find invertible matrices A and B such that $A + B$ is not invertible.**
    (b) **Find singular matrices A and B such that $A + B$ is invertible.**

12. **If the product $C = AB$ is invertible (A and B are square), then A itself is invertible. Find a formula for A^{-1} that involves C^{-1} and B.**

13. **If the product $M = ABC$ of three square matrices is invertible, then B is invertible. (So are A and C.) Find a formula for B^{-1} that involves M^{-1} and A and C.**

14. **If you add row 1 of A to row 2 to get B, how do you find B^{-1} from A^{-1}?**
    Notice the order. The inverse of $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} A$ is ________.

15. **Prove that a matrix with a column of zeros cannot have an inverse.**

16. **Multiply $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ times $\begin{bmatrix} -d & -b \\ c & a \end{bmatrix}$. What is the inverse of each matrix if $ad \neq bc$?**

17. (a) **What 3 by 3 matrix E has the same effect as these three steps? Subtract row 1 from row 2, subtract row 1 from row 3, then subtract row 2 from row 3.**
    (b) **What single matrix L has the same effect as these three reverse steps? Add row 2 to row 3, add row 1 to row 3.**

18. **If B is the inverse of A^2, show that AB is the inverse of A.**

19. **Find the numbers a and b that give the inverse of $5 \cdot \textbf{eye}(4) - \textbf{ones}(4, 4)$:**

$$\begin{bmatrix} -1 & 4 & -1 & -1 \\ -1 & 4 & -1 & -1 \\ -1 & -1 & 4 & -1 \\ -1 & -1 & -1 & 4 \end{bmatrix}^{-1} = \begin{bmatrix} a & b & b & b \\ b & a & b & b \\ b & b & a & b \\ b & b & b & a \end{bmatrix}.$$

    What are a and b in the inverse of $6 \cdot \textbf{eye}(5) - \textbf{ones}(5, 5)$?

20. **Show that $A = 4 \cdot \textbf{eye}(4) - \textbf{ones}(4, 4)$ is not invertible: Multiply $A \cdot \textbf{ones}(4, 1)$.**

21. **There are sixteen 2 by 2 matrices whose entries are 1's and 0's. How many of them are invertible?**

Questions 22–28 are about the Gauss-Jordan method for calculating A^{-1}.

22. Change I into A^{-1} as you reduce A to I (by row operations):

$$[A \mid I] = \begin{bmatrix} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{bmatrix} \quad \textbf{and} \quad [A \mid I] = \begin{bmatrix} 1 & 4 & 1 & 0 \\ 3 & 9 & 0 & 1 \end{bmatrix}.$$

23. Follow the 3 by 3 text example but with plus signs in A. Eliminate above and below the pivots to reduce $[A \mid I]$ to $[I \mid A^{-1}]$:

$$[A \mid I] = \begin{bmatrix} 2 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{bmatrix}.$$

24. Use Gauss-Jordan elimination on $[U \mid I]$ to find the upper triangular U^{-1}:

$$UU^{-1} = I \qquad \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

25. Find A^{-1} and B^{-1} (if they exist) by elimination on $[A \mid I]$ and $[B \mid I]$:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \quad \textbf{and} \quad B = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}.$$

26. What three matrices E_{21} and E_{12} and D^{-1} reduce $A = \begin{bmatrix} 1 & 2 \\ 2 & 6 \end{bmatrix}$ to the identity matrix? Multiply $D^{-1}E_{12}E_{21}$ to find A^{-1}.

27. Invert these matrices A by the Gauss-Jordan method starting with $[A \mid I]$:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \quad \textbf{and} \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}.$$

28. Exchange rows and continue with Gauss-Jordan to find A^{-1}:

$$[A \mid I] = \begin{bmatrix} 0 & 2 & 1 & 0 \\ 2 & 2 & 0 & 1 \end{bmatrix}.$$

29. True or false (with a counterexample if false and a reason if true):

- **(a) A 4 by 4 matrix with a row of zeros is not invertible.**
- **(b) Every matrix with 1's down the main diagonal is invertible.**
- **(c) If A is invertible then A^{-1} and A^2 are invertible.**

30. (Recommended) Prove that A is invertible if $a \neq 0$ and $a \neq b$ (find the pivots or A^{-1}). Then find three numbers c so that C is not invertible:

$$A = \begin{bmatrix} a & b & b \\ a & a & b \\ a & a & a \end{bmatrix} \quad \textbf{and} \quad C = \begin{bmatrix} 2 & c & c \\ c & c & c \\ 8 & 7 & c \end{bmatrix}.$$

31. This matrix has a remarkable inverse. Find A^{-1} by elimination on $[A \mid I]$. Extend to a 5 by 5 "alternating matrix" and guess its inverse; then multiply to confirm.

Invert $A = \begin{bmatrix} 1 & -1 & 1 & -1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ **and solve** $A\boldsymbol{x} = (1, 1, 1, 1)$.

32. Suppose the matrices P and Q have the same rows as I but in any order. They are "permutation matrices". Show that $P - Q$ is singular by solving $(P - Q)\boldsymbol{x} = \mathbf{0}$.

33. Find and check the inverses (assuming they exist) of these block matrices:

$$\begin{bmatrix} I & 0 \\ C & I \end{bmatrix} \quad \begin{bmatrix} A & 0 \\ C & D \end{bmatrix} \quad \begin{bmatrix} 0 & I \\ I & D \end{bmatrix}.$$

34. Could a 4 by 4 matrix A be invertible if every row contains the numbers $0, 1, 2, 3$ in some order? What if every row of B contains $0, 1, 2, -3$ in some order?

35. In the Worked Example 2.5 C, the triangular Pascal matrix L has $L^{-1} = LDL$, where the diagonal matrix D has alternating entries $1, -1, 1, -1$. Then $LDLD = I$, so the inverse of $LD =$ pascal$(4, 1)$.
So what is the inverse of $LD =$ pascal$(4, 1)$?

36. The Hilbert matrices have $H_{ij} = 1/(i+j-1)$. Ask MATLAB for the exact 6 by 6 inverse inv$(H(6))$. Then ask for it to compute inv$(H(6))$. How can these be different, when the computer never makes mistakes?

37. (a) Use inv(P) to invert MATLAB's 4 by 4 symmetric matrix $P =$ pascal(4).
(b) Create Pascal's lower triangular $L =$ abs(pascal$(4, 1)$) and test $P = LL^T$.

38. If $A =$ ones(4) and $b =$ rand$(4, 1)$, how does MATLAB tell you that $Ax = \boldsymbol{b}$ has no solution? For the special $b =$ ones$(4, 1)$, which solution to $Ax = \boldsymbol{b}$ is found by $A\backslash b$?

Challenge Problems

39. (Recommended) A is a 4 by 4 matrix with 1 's on the diagonal and -a, -b, -c on the diagonal above. Find A^{-1} for this bidiagonal matrix.

40. Suppose E_1, E_2, E_3 are 4 by 4 identity matrices, except E_1 has a, b, c in column 1 and E_2 has d, e in column 2 and E_3 has f in column 3 (below the 1's). Multiply $L = E_1E_2E_3$ to show that all these nonzeros are copied into L.
$E_1E_2E_3$ is in the *opposite* order for elimination (because E_3 is acting first). But $E_1E_2E_3 = L$ is in the *correct* order for invert elimination and recover A.

41. Second difference matrices have beautiful inverses if they start with $T_{11} = 1$ (instead of $K_{11} = 2$). Here is the 3 by 3 tridiagonal matrix T and its inverse:

$$T = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \qquad T^{-1} = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

One approach is Gauss-Jordan elimination on $[T \mid I]$. I would rather write T as the product of first differences L times U. The inverses of L and U in Worked Example 2.5 A are sum matrices, so here are $T = LU$ and $T^{-1} = U^{-1}L^{-1}$:

$$T = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \qquad T^{-1} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Question. (4 by 4) What are the pivots of T? What is its 4 by 4 inverse? The reverse order UL gives what matrix T^{**}? What is the inverse of T^{**}?

42. Here are two more difference matrices, both important. But are they invertible?

$$\textbf{Cyclic } C = \begin{bmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix} \qquad \textbf{Free ends } F = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

43. Elimination for a block matrix: When you multiply the first block row $[A\ B]$ by CA^{-1} and subtract from the second row $[C\ D]$, the "Schur complement" S appears:

$$\begin{bmatrix} I & 0 \\ -CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & S \end{bmatrix}, \qquad \textbf{where } A \textbf{ and } D \textbf{ are square}, \quad S = D - CA^{-1}B.$$

Multiply on the right to subtract $A^{-1}B$ times block column 1 from block column 2:

$$\begin{bmatrix} A & B \\ 0 & S \end{bmatrix} \begin{bmatrix} I & -A^{-1}B \\ 0 & I \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & S \end{bmatrix}.$$

Find S for

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 2 & 3 & 3 \\ 4 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}.$$

The block pivots are A and S. If they are invertible, so is $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$.

44. **How does the identity $A(I+BA) = (I+AB)A$ connect the inverses of $I+BA$ and $I+AB$? Those are both invertible or both singular: not obvious.**

2.6 Elimination = Factorization: $A = LU$

1. **(Important) Forward elimination changes $\begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix} \boldsymbol{x} = \boldsymbol{b}$ to a triangular $\begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix} \boldsymbol{x} = \boldsymbol{c}$:**

$$\begin{aligned} x + y &= 5 \\ x + 2y &= 7 \end{aligned} \quad \longrightarrow \quad \begin{aligned} x + y &= 5 \\ y &= 2 \end{aligned} \qquad \begin{bmatrix} 1 & 1 & 5 \\ 1 & 2 & 7 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & 5 \\ 0 & 1 & 2 \end{bmatrix}.$$

That step subtracted $\ell_{21} =$ ____ times row 1 from row 2. The reverse step adds ℓ_{21} times row 1 to row 2. The matrix for that reverse step is $L =$ ________. Multiply this L times the triangular system $\begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix} \boldsymbol{x}_1 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$ to get ________ $=$ ________. In letters, L multiplies $U\boldsymbol{x} = \boldsymbol{c}$ to give ________.

2. **Write down the 2 by 2 triangular systems $L\boldsymbol{c} = \boldsymbol{b}$ and $U\boldsymbol{x} = \boldsymbol{c}$ from Problem 1. Check that $\boldsymbol{c} = (5, 2)$ solves the first one. Find $\boldsymbol{x}$ that solves the second one.**

3. **(Move to 3 by 3) Forward elimination changes $A\boldsymbol{x} = \boldsymbol{b}$ to a triangular $U\boldsymbol{x} = \boldsymbol{c}$:**

$$\begin{aligned} x + y + z &= 5 \\ x + y + 2z &= 7 \\ x + 3y + 6z &= 11 \end{aligned} \quad \longrightarrow \quad \begin{aligned} x + y + z &= 5 \\ y + 2z &= 2 \\ 2y + 5z &= 6 \end{aligned} \quad \longrightarrow \quad \begin{aligned} x + y + z &= 5 \\ y + 2z &= 2 \\ z &= 2 \end{aligned}$$

The equation $z = 2$ in $U\boldsymbol{x} = \boldsymbol{c}$ comes from the original $x + 3y + 6z = 11$ in $A\boldsymbol{x} = \boldsymbol{b}$ by subtracting $\ell_{31} =$ ____ times equation 1 and $\ell_{32} =$ ____ times the final equation 2. Reverse that to recover $\begin{bmatrix} 1 & 3 & 6 \\ 1 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ in the last row of A and $\boldsymbol{b}$ from the final $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 5 \\ 2 \\ 2 \end{bmatrix}$ in U and $\boldsymbol{c}$:
Row 3 of $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix} = (\ell_{31}$ Row $1 + \ell_{32}$ Row $2 + 1$ Row $3)$ of $\begin{bmatrix} U & \boldsymbol{c} \end{bmatrix}$.
In matrix notation this is multiplication by L. So $A = LU$ and $\boldsymbol{b} = L\boldsymbol{c}$.

4. **What are the 3 by 3 triangular systems $L\boldsymbol{c} = \boldsymbol{b}$ and $U\boldsymbol{x} = \boldsymbol{c}$ from Problem 3? Check that $\boldsymbol{c} = (5, 2, 2)$ solves the first one. Which $\boldsymbol{x}$ solves the second one?**

5. **What matrix E puts A into triangular form $EA = U$? Multiply by $E^{-1} = L$ to factor A into LU:**

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 2 \\ 6 & 3 & 5 \end{bmatrix}.$$

6. **What two elimination matrices E_{21} and E_{32} put A into upper triangular form $E_{32}E_{21}A = U$? Multiply by E_{21}^{-1} and E_{32}^{-1} to factor A into $LU = E_{21}^{-1}E_{32}^{-1}U$:**

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 5 \\ 0 & 4 & 0 \end{bmatrix}.$$

7. **What three elimination matrices E_{21}, E_{31}, E_{32} put A into its upper triangular form $E_{32}E_{31}E_{21}A = U$? Multiply by $E_{21}^{-1}, E_{31}^{-1}, E_{32}^{-1}$ to factor A into L times U:**

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 2 & 2 \\ 3 & 4 & 5 \end{bmatrix}, \qquad L = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}.$$

8. **This is the problem that shows how the inverses E_{ij}^{-1} multiply to give L. You see this best when A is already lower triangular with 1's on the diagonal. Then $U = I$!**

$$A = L = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}.$$

The elimination matrices E_{21}, E_{31}, E_{32} contain $-a$ then $-b$ then $-c$.

   (a) Multiply $E_{32}E_{31}E_{21}$ to find the single matrix E that produces $EA = I$.

   (b) Multiply $E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}$ to bring back L. The multipliers a, b, c are mixed up in E but perfect in L.

9. **When zero appears in a pivot position, $A = LU$ is not possible! (We are requiring nonzero zeros in U.) Show directly why these equations are both impossible:**

$$\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \ell & 1 \end{bmatrix} \begin{bmatrix} d & e \\ 0 & f \end{bmatrix} \qquad \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \ell & 1 & 0 \\ m & n & 1 \end{bmatrix} \begin{bmatrix} d & e & g \\ 0 & f & h \\ 0 & 0 & i \end{bmatrix}.$$

These matrices need a row exchange. That uses a "permutation matrix" P.

10. **Which number c leads to zero in the second pivot position? A row exchange is needed and $A = LU$ will not be possible. Which c produces zero in the third pivot position? Then a row exchange can't help and elimination fails:**

$$A = \begin{bmatrix} 1 & c & 0 \\ 2 & 4 & 1 \\ 3 & 5 & 1 \end{bmatrix}.$$

11. **What are L and D (the diagonal pivot matrix) for this matrix A? What is U in $A = LU$ and what is the new U in $A = LDU$?**

$$\textbf{Already triangular} \qquad A = \begin{bmatrix} 2 & 4 & 8 \\ 0 & 3 & 9 \\ 0 & 0 & 7 \end{bmatrix}.$$

12. **A and B are symmetric across the diagonal (for $4 = 4$). Find their triple factorizations LDU and say how U is related to L for these symmetric matrices:**

$$\textbf{Symmetric} \qquad A = \begin{bmatrix} 2 & 4 \\ 4 & 11 \end{bmatrix} \quad \textbf{and} \quad B = \begin{bmatrix} 1 & 4 & 0 \\ 4 & 12 & 4 \\ 0 & 4 & 0 \end{bmatrix}.$$

13. **(Recommended) Compute L and U for the symmetric matrix A:**

$$A = \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix}.$$

Find four conditions on a, b, c, d to get $A = LU$ with four pivots.

14. This nonsymmetric matrix will have the same L as in Problem 13:

$$\textbf{Find } L \textbf{ and } U \textbf{ for} \qquad A = \begin{bmatrix} a & r & r & r \\ a & b & s & s \\ a & b & c & t \\ a & b & c & d \end{bmatrix}.$$

Find the four conditions on a, b, c, d, r, s, t to get $A = LU$ with four pivots.

Problems 15–16 use L and U (without needing A) to solve $A\boldsymbol{x} = \boldsymbol{b}$.

15. Solve the triangular system $L\boldsymbol{c} = \boldsymbol{b}$ to find $\boldsymbol{c}$. Then solve $U\boldsymbol{x} = \boldsymbol{c}$ to find $\boldsymbol{x}$:

$$L = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 2 & 4 \\ 0 & 1 \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} 2 \\ 11 \end{bmatrix}.$$

For safety multiply LU and solve $A\boldsymbol{x} = \boldsymbol{b}$ as usual. Circle $\boldsymbol{c}$ when you see it.

16. Solve $L\boldsymbol{c} = \boldsymbol{b}$. Then solve $U\boldsymbol{x} = \boldsymbol{c}$. What was A?

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}.$$

17. (a) When you apply the usual elimination steps to L, what matrix do you reach?

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix}.$$

(b) When you apply the same steps to I, what matrix do you get?

(c) When you apply the same steps to LU, what matrix do you get?

18. If $A = LDU$ and also $A = L_1 D_1 U_1$ with all factors invertible, then $L = L_1$ and $D = D_1$ and $U = U_1$. *"The three factors are unique."*
Derive the equation $L_1^{-1} L D = D_1 U_1 U^{-1}$. Are the two sides triangular or diagonal? Deduce $L = L_1$ and $U = U_1$ (they all have diagonal 1's). Then $D = D_1$.

19. Tridiagonal matrices have zero entries except on the main diagonal and the two adjacent diagonals. Factor these into $A = LU$ and $A = LDL^T$:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \quad \textbf{and} \quad A = \begin{bmatrix} a & a & 0 \\ a & a+b & b \\ 0 & b & b+c \end{bmatrix}.$$

20. When A is tridiagonal, its L and U factors have only two nonzero diagonals. How would you take advantage of knowing the zeros in T, in a code for Gaussian elimination? Find L and U.

$$\textbf{Tridiagonal} \qquad T = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 2 & 3 & 1 & 0 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 3 & 4 \end{bmatrix}.$$

21. If A and B have nonzeros in the positions marked by x, which zeros (marked by 0) stay zero in their factors L and U?

$$A = \begin{bmatrix} x & x & x & x \\ x & x & x & 0 \\ 0 & x & x & x \\ 0 & 0 & x & x \end{bmatrix} \qquad B = \begin{bmatrix} x & x & x & 0 \\ x & x & 0 & x \\ x & 0 & x & x \\ 0 & x & x & x \end{bmatrix}.$$

22. Suppose you eliminate upwards (almost unheard of). Use the last row to produce zeros in the last column (the pivot is 1). Then use the second row to produce zero above the second pivot. Find the factors in the unusual order $A = UL$.

$$\textbf{Upper times lower} \qquad A = \begin{bmatrix} 5 & 3 & 1 \\ 3 & 3 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

23. Easy but important. If A has pivots $5, 9, 3$ with no row exchanges, what are the pivots for the upper left 2 by 2 submatrix A_2 (without row 3 and column 3)?

Challenge Problems

24. **Which invertible matrices allow $A = LU$ (elimination without row exchanges)? *Good question!* Look at each of the square upper left submatrices A_k of A.**

    All upper left k by k submatrices A_k must be invertible(sizes $k = 1, \ldots,$ n).

    Explain that answer: A_k factors into ________ because $LU = \begin{bmatrix} L_k & 0 \\ * & * \end{bmatrix} \begin{bmatrix} U_k & * \\ 0 & * \end{bmatrix}$.

25. **For the 6 by 6 second difference constant-diagonal matrix K, put the pivots and multipliers into $K = LU$. (L and U will have only two nonzero diagonals, because K has three.) Find a formula for the i, j entry of L^{-1}, by software like MATLAB using inv(L) or by looking for a nice pattern.**

$$-1, 2, -1 \textbf{ matrix} \quad K = \begin{bmatrix} 2 & -1 & & & & \\ -1 & 2 & -1 & & & \\ & \ddots & \ddots & \ddots & & \\ & & -1 & 2 & -1 & \\ & & & -1 & 2 & -1 \\ & & & & -1 & 2 \end{bmatrix} = \textbf{toeplitz}([2 \ \ -1 \ 0 \ 0 \ 0 \ 0]).$$

26. **If you print K^{-1}, it doesn't look so good (6 by 6). But if you print $7K^{-1}$, that matrix looks wonderful. Write down $7K^{-1}$ by hand, following this pattern:**

    1. **Row 1 and column 1 are $(6, 5, 4, 3, 2, 1)$.**
    2. **On and above the main diagonal, row i is i times row 1.**
    3. **On and below the main diagonal, column j is j times column 1.**

    Multiply K times that $7K^{-1}$ to produce $7I$. Here is $4K^{-1}$ for $n = 3$:

$$\begin{array}{l} \textit{3 by 3 case} \\ \textit{The determinant} \\ \textit{of this } K \textit{ is } 4. \end{array} \qquad (K)(4K^{-1}) = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 4 & & \\ & 4 & \\ & & 4 \end{bmatrix}.$$

2.7 Transposes and Permutations

Questions 1–7 are about the rules for transpose matrices.

1. **Find A^{T} and A^{-1} and $(A^{-1})^{\mathrm{T}}$ and $(A^{\mathrm{T}})^{-1}$ for**

$$A = \begin{bmatrix} 1 & 0 \\ 9 & 3 \end{bmatrix} \quad \textbf{and also} \quad A = \begin{bmatrix} 1 & c \\ c & 0 \end{bmatrix}.$$

2. **Verify that $(AB)^{\mathrm{T}}$ equals $B^{\mathrm{T}}A^{\mathrm{T}}$ but those are different from $A^{\mathrm{T}}B^{\mathrm{T}}$:**

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}, \quad AB = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}.$$

    Show also that AA^{T} is different from $A^{\mathrm{T}}A$. But both of those matrices are ___________.

3. **(a) The matrix $((AB)^{-1})^{\mathrm{T}}$ comes from $(A^{-1})^{\mathrm{T}}$ and $(B^{-1})^{\mathrm{T}}$. In what order?**

    (b) If U is upper triangular then $(U^{-1})^{\mathrm{T}}$ is ___________ triangular.

4. **Show that $A^2 = 0$ is possible but $A^{\mathrm{T}}A = 0$ is not possible (unless $A =$ zero matrix).**

5. **(a) The row vector x^{T} times A times the column y produces what number?**

$$x^{\mathrm{T}}Ay = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \text{\_\_\_\_\_\_\_\_\_\_\_}.$$

    (b) This is the row $x^{\mathrm{T}}A =$ ___________ times the column $y = (0, 1, 0)$.

(c) **This is the row $x^{\mathrm{T}} = \begin{bmatrix} 0 & 1 \end{bmatrix}$ times the column $Ay =$ ____________.**

6. **The transpose of a block matrix $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ is $M^{\mathrm{T}} =$ ____________. Test an example. Under what conditions on A, B, C, D is the block matrix symmetric?**

7. **True or false:**

   (a) **The block matrix $\begin{bmatrix} 0 & A \\ A & 0 \end{bmatrix}$ is automatically symmetric.**

   (b) **If A and B are symmetric then their product AB is symmetric.**

   (c) **If A is not symmetric then A^{-1} is not symmetric.**

   (d) **When A, B, C are symmetric, the transpose of ABC is CBA.**

Questions 8–15 are about permutation matrices.

8. **Why are there $n!$ permutation matrices of order n?**

9. **If P_1 and P_2 are permutation matrices, so is P_1P_2. This still has the rows of I in some order. Give examples with $P_1P_2 \neq P_2P_1$ and $P_3P_4 = P_4P_3$.**

10. **There are 12 "even" permutations of $(1, 2, 3, 4)$, with an even number of exchanges. Two of them are $(1, 2, 3, 4)$ with no exchanges and $(4, 3, 2, 1)$ with two exchanges. List the other ten. Instead of writing 4 by 4 matrix, just order the numbers.**

11. **Which permutation makes PA upper triangular? Which permutations make P_1AP_2 lower triangular? Multiplying A on the right by P_2 exchanges the ____________ of A.**

$$A = \begin{bmatrix} 0 & 0 & 6 \\ 1 & 2 & 3 \\ 0 & 4 & 5 \end{bmatrix}.$$

12. **Explain why the dot product of x and y equals the dot product of Px and Py. Then $(Px)^{\mathrm{T}}(Py) = x^{\mathrm{T}}y$ tells us that $P^{\mathrm{T}}P = I$ for any permutation. With $x = (1, 2, 3)$ and $y = (1, 4, 2)$ choose P to show that $Px \cdot Py$ is not always $x \cdot y$.**

13. (a) **Find a 3 by 3 permutation matrix with $P^3 = I$ (but not $P = I$).**

    (b) **Find a 4 by 4 permutation $\bar{P}$ with $\bar{P}^4 \neq I$.**

14. **If P has 1's on the antidiagonal from $(1, n)$ to $(n, 1)$, describe PAP. Note $P = P^{\mathrm{T}}$.**

15. **All row exchange matrices are symmetric: $P^{\mathrm{T}} = P$. Then $P^{\mathrm{T}}P = I$ becomes $P^2 = I$. Other permutation matrices may or may not be symmetric.**

    (a) **If P sends row 1 to row 4, then P^{T} sends row ____________ to row ____________. When $P^{\mathrm{T}} = P$ the row exchanges come in pairs with no overlap.**

    (b) **Find a 4 by 4 example with $P^{\mathrm{T}} = P$ that moves all four rows.**

Questions 16–21 are about symmetric matrices and their factorizations.

16. **If $A = A^{\mathrm{T}}$ and $B = B^{\mathrm{T}}$, which of these matrices are certainly symmetric?**

    (a) $A^2 - B^2$

    (b) $(A + B)(A - B)$

    (c) ABA

    (d) $ABAB$

17. **Find 2 by 2 symmetric matrices $S = S^{\mathrm{T}}$ with these properties:**

    (a) **S is not invertible.**

    (b) **S is invertible but cannot be factored into LU (row exchanges needed).**

    (c) **S can be factored into LDL^{T} but not into LL^{T} (because of negative D).**

18. **(a) How many entries of S can be chosen independently, if $S = S^{\mathrm{T}}$ is 5 by 5?**

(b) How do L and D (still 5 by 5) give the same number of choices in LDL^{T}?

(c) How many entries can be chosen if A is skew-symmetric? ($A^{\mathrm{T}} = -A$).

19. **Suppose A is rectangular (m by n) and S is symmetric (m by m).**

(a) Transpose $A^{\mathrm{T}}SA$ to show its symmetry. What shape is this matrix?

(b) Show why $A^{\mathrm{T}}A$ has no negative numbers on its diagonal.

20. **Factor these symmetric matrices into $S = LDL^{\mathrm{T}}$. The pivot matrix D is diagonal:**

$$S = \begin{bmatrix} 1 & 3 \\ 3 & 2 \end{bmatrix}, \quad S = \begin{bmatrix} 1 & b \\ b & c \end{bmatrix}, \quad S = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}.$$

21. **After elimination clears out column 1 below the first pivot, find the symmetric 2 by 2 matrix that appears in the lower right corner:**

$$\textbf{Start from } S = \begin{bmatrix} 2 & 4 & 8 \\ 4 & 3 & 9 \\ 8 & 9 & 0 \end{bmatrix} \quad \textbf{and} \quad S = \begin{bmatrix} 1 & b & c \\ b & d & e \\ c & e & f \end{bmatrix}.$$

Questions 22–24 are about the factorizations $PA = LU$ and $A = L_1P_1U_1$.

22. **Find the $PA = LU$ factorizations (and check them) for**

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

23. **Find a 4 by 4 permutation matrix (call it A) that needs 3 row exchanges to reach the end of elimination. For this matrix, what are its factors $P, L,$ and U?**

24. **Factor the following matrix into $PA = LU$. Factor it also into $A = L_1P_1U_1$ (hold the exchange of row 3 until 3 times row 1 is subtracted from row 2):**

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 3 & 8 \\ 2 & 1 & 1 \end{bmatrix}.$$

25. **Prove that the identity matrix cannot be the product of three row exchanges (or five). It can be the product of two exchanges (or four).**

26. **(a) Choose E_{21} to remove the 3 below the first pivot. Then multiply $E_{21}SE_{21}^{\mathrm{T}}$ to remove both 3's:**

$$S = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 11 & 4 \\ 0 & 4 & 9 \end{bmatrix} \quad \textbf{is going toward} \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(b) Choose E_{32} to remove the 4 below the second pivot. Then S is reduced to D by $E_{32}E_{21}SE_{21}^{\mathrm{T}}E_{32}^{\mathrm{T}} = D$. Invert the E's to find L in $S = LDL^{\mathrm{T}}$.

27. **If every row of a 4 by 4 matrix contains the numbers $0, 1, 2, 3$ in some order, can the matrix be symmetric?**

28. **Prove that no reordering of rows and reordering of columns can transpose a typical matrix. (Watch the diagonal entries.)**

The next three questions are about applications of the identity $(Ax)^{\mathrm{T}}y = x^{\mathrm{T}}(A^{\mathrm{T}}y)$.

29. Wires go between Boston, Chicago, and Seattle. Those cities are at voltages x_B, x_C, x_S. With unit resistances between cities, the currents between cities are in y:

$$y = Ax \quad \textbf{is} \quad \begin{bmatrix} y_{BC} \\ y_{CS} \\ y_{BS} \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_B \\ x_C \\ x_S \end{bmatrix}.$$

(a) Find the total currents $A^{\mathrm{T}}y$ out of the three cities.

(b) Verify that $(Ax)^{\mathrm{T}}y$ agrees with $x^{\mathrm{T}}(A^{\mathrm{T}}y)$—six terms in both.

30. Producing x_1 trucks and x_2 planes needs $x_1 + 50x_2$ tons of steel, $40x_1 + 1000x_2$ pounds of rubber, and $2x_1 + 50x_2$ months of labor. If the unit costs y_1, y_2, y_3 are $700 per ton, $3 per pound, and $3000 per month, what are the values of one truck and one plane? Those are the components of $A^{\mathrm{T}}y$.

31. Ax gives the amounts of steel, rubber, and labor to produce x in Problem 31. Find A. Then $Ax \cdot y$ is the ____________ of inputs while $x \cdot A^{\mathrm{T}}y$ is the value of ____________.

32. The matrix P that multiplies (x, y, z) to give (z, x, y) is also a rotation matrix. Find P and P^3. The rotation axis $a = (1, 1, 1)$ doesn't move, it equals Pa. What is the angle of rotation from $v = (2, 3, -5)$ to $Pv = (-5, 2, 3)$?

33. Write $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$ as the product ES of an elementary row operation matrix E and a symmetric matrix S.

34. Here is a new factorization of A into LS: triangular (with 1's) times symmetric:

Start from $A = LDU$. Then A equals $L(U^{\mathrm{T}})^{-1}$ times $S = U^{\mathrm{T}}DU$.

Why is $L(U^{\mathrm{T}})^{-1}$ triangular? Its diagonal is all 1's. Why is $U^{\mathrm{T}}DU$ symmetric?

35. A group of matrices includes AB and A^{-1} if it includes A and B. "Products and inverses stay in the group." Which of these sets are groups?
Lower triangular matrices L with 1's on the diagonal, symmetric matrices S, positive matrices M, diagonal invertible matrices D, permutation matrices P, matrices with $Q^{\mathrm{T}} = Q^{-1}$. Invent two more matrix groups.

Challenge Problems

36. A square northwest matrix B is zero in the southeast corner, below the antidiagonal that connects $(1, n)$ to $(n, 1)$. Will B^{T} and B^2 be northwest matrices? Will B^{-1} be northwest or southeast? What is the shape of BC = northwest times southeast?

37. If you take powers of a permutation matrix P, why is some P^k eventually equal to I? Find a 5 by 5 permutation P so that the smallest power to equal I is P^6.

38. (a) Write down any 3 by 3 matrix M. Split M into $S + A$ where $S = S^{\mathrm{T}}$ is symmetric and $A = -A^{\mathrm{T}}$ is anti-symmetric.

(b) Find formulas for S and M^{T}. We want $M = S + A$.

39. Suppose Q^{T} equals Q^{-1} (transpose equals inverse, so $Q^{\mathrm{T}}Q = I$).

(a) Show that the columns $q_1, \ldots, q_n$ are unit vectors: $\|q_i\|^2 = 1$.

(b) Show that every two columns of Q are perpendicular: $q_1^{\mathrm{T}}q_2 = 0$.

(c) Find a 2 by 2 example with first entry $q_{11} = \cos\theta$.

3 Chapter 3: Vector Spaces and Subspaces

3.1 Spaces of Vectors

The first problems 1–8 are about vector spaces in general. The vectors in those spaces are not necessarily column vectors. In the definition of a *vector space*, vector addition $x + y$ and scalar multiplication cx must obey the following eight rules:

(1) $x + y = y + x$

(2) $x + (y + z) = (x + y) + z$

(3) There is a unique "zero vector" such that $x + 0 = x$ **for all** x

(4) For each x **there is a unique vector** $-x$ **such that** $x + (-x) = 0$

(5) 1 **times** x **equals** x

(6) $(c_1 c_2)x = c_1(c_2 x)$

(7) $c(x + y) = cx + cy$

(8) $(c_1 + c_2)x = c_1 x + c_2 x$

(1) to (4) about $x + y$ **(5) to (6) about** cx **(7) to (8) connects them**

1. **Suppose** $(x_1, x_2) + (y_1, y_2)$ **is defined to be** $(x_1 + y_2, x_2 + y_1)$**. With the usual multiplication** $cx = (cx_1, cx_2)$**, which of the eight conditions are not satisfied?**

2. **Suppose the multiplication** cx **is defined to produce** $(cx_1, 0)$ **instead of** (cx_1, cx_2)**. With the usual addition in** $\mathbb{R}^2$**, are the eight conditions satisfied?**

3. **(a) Which rules are broken if we keep only the positive numbers** $x > 0$ **in** $\mathbb{R}^1$**? Every** c **must be allowed. The half-line is not a subspace.**

   (b) The positive numbers with $x + y$ **and** cx **redefined to equal the usual** xy **and** x^c **do satisfy the eight rules. Test rule 7 when** $c = 3$, $x = 2$, $y = 1$**. (Then** $x + y = 2$ **and** $cx = 8$**.) Which number acts as the "zero vector"?**

4. **The matrix** $A = \begin{bmatrix} 2 & -2 \\ 2 & -2 \end{bmatrix}$ **is a "vector" in the space M of all 2 by 2 matrices. Write down the zero vector in this space, the vector** $\frac{1}{2}A$**, and the vector** $-A$**. What matrices are in the smallest subspace containing** A**?**

5. **Describe a subspace of M that contains** $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ **but not** $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$**.**

   (a) If a subspace of M contains A **and** B**, must it contain** 0**?**

   (b) Describe a subspace of M that contains no nonzero diagonal matrices.

6. **The functions** $f(x) = x^2$ **and** $g(x) = 5x$ **are "vectors" in F. This is the vector space of all real functions. (The functions are defined for** $-\infty < x < \infty$**.) The combination** $3f(x) - 4g(x)$ **is the function** $h(x) =$ __________________**.**

7. **Which rule is broken if multiplying** $f(x)$ **by** c **gives the function** $f(cx)$**? Keep the usual addition,** $f(x) + g(x)$**.**

8. **If the sum of the "vectors"** $f(x)$ **and** $g(x)$ **is defined to be the function** $f(g(x))$**, then the "zero vector" is** $g(x) = x$**. Keep the usual scalar multiplication** $cf(x)$ **and find two rules that are broken.**

Questions 9–18 are about the "subspace requirements": $x + y$ **and** cx **and then all linear combinations** $cx + dy$ **stay in the subspace.**

9. **One requirement can be met while the other fails. Show this by finding**

   (a) A set of vectors in $\mathbb{R}^2$ **for which** $x + y$ **stays in the set but** $\frac{1}{2}x$ **may be outside.**

(b) **A set of vectors in $\mathbb{R}^2$ (other than two quarter-planes) for which every cx stays in the set but $x + y$ may be outside.**

10. **Which of the following subsets of $\mathbb{R}^3$ are actually subspaces?**

    (a) **The plane of vectors (b_1, b_2, b_3) with $b_1 = b_2$.**

    (b) **The plane of vectors with $b_1 = 1$.**

    (c) **The vectors with $b_1 b_2 b_3 = 0$.**

    (d) **All linear combinations of $v = (1, 4, 0)$ and $w = (2, 2, 2)$.**

    (e) **All vectors that satisfy $b_1 + b_2 + b_3 = 0$.**

    (f) **All vectors with $b_1 \leq b_2 \leq b_3$.**

11. **Describe the smallest subspace of the matrix space M that contains**

    (a) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ **and** $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

    (b) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

    (c) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ **and** $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

12. **Let P be the plane in $\mathbb{R}^3$ with equation $x + y - 2z = 4$. The origin $(0, 0, 0)$ is not in P! Find two vectors in P and check that their sum is not in P.**

13. **Let P_0 be the plane through $(0, 0, 0)$ parallel to the previous plane P. What is the equation for P_0? Find two vectors in P_0 and check that their sum is in P_0.**

14. **The subspaces of $\mathbb{R}^3$ are planes, lines, $\mathbb{R}^3$ itself, or $\mathbb{Z}$ containing only $(0, 0, 0)$.**

    (a) **Describe the three types of subspaces of $\mathbb{R}^2$.**

    (b) **Describe all subspaces of D, the space of 2 by 2 diagonal matrices.**

15. (a) **The intersection of two planes through $(0, 0, 0)$ is probably a ____________ in $\mathbb{R}^3$ but it could be a ____________. It can't be $\mathbb{Z}$!**

    (b) **The intersection of a plane through $(0, 0, 0)$ with a line through $(0, 0, 0)$ is probably a ____________ but it could be a ____________.**

    (c) **If S and T are subspaces of $\mathbb{R}^5$, prove that their intersection S $\cap$ T is a subspace of $\mathbb{R}^5$. Here S $\cap$ T consists of the vectors that lie in both subspaces. Check that $x + y$ and cx are in S $\cap$ T if x and y are in both spaces.**

16. **Suppose P is a plane through $(0, 0, 0)$ and L is a line through $(0, 0, 0)$. The smallest vector space containing both P and L is either ____________ or ____________.**

17. (a) **Show that the set of invertible matrices in M is not a subspace.**

    (b) **Show that the set of singular matrices in M is not a subspace.**

18. **True or false (check addition in each case by example):**

    (a) **The symmetric matrices in M (with $A^{\mathrm{T}} = A$) form a subspace.**

    (b) **The skew-symmetric matrices in M (with $A^{\mathrm{T}} = -A$) form a subspace.**

    (c) **The unsymmetric matrices in M (with $A^{\mathrm{T}} \neq A$) form a subspace.**

Questions 19–27 are about column spaces $C(A)$ and the equation $Ax = b$.

19. **Describe the column spaces (lines or planes) of these particular matrices:**

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 0 \end{bmatrix}.$$

20. For which right sides (find a condition on b_1, b_2, b_3) are these systems solvable?

(a) $\begin{bmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ -1 & -4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 4 \\ 2 & 9 \\ -1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

21. Adding row 1 of A to row 2 produces B. Adding column 1 to column 2 produces C. A combination of the columns of (B or C?) is also a combination of the columns of A. Which two matrices have the same column space?

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}.$$

22. For which vectors (b_1, b_2, b_3) do these systems have a solution?

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad \textbf{and} \quad \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

23. (Recommended) If we add an extra column b to a matrix A, then the column space gets larger unless ____________________. Give an example where the column space gets larger and an example where it doesn't. Why is $Ax = b$ solvable exactly when the column space doesn't get larger—it is the same for A and $[A\ b]$?

24. The columns of AB are combinations of the columns of A. This means: *The column space of* AB *is contained in* (possibly equal to) *the column space of* A. Give an example where the column spaces of A and AB are not equal.

25. Suppose $Ax = b$ and $Ay = b^*$ are both solvable. Then $Az = b + b^*$ is solvable. What is z? This translates into: If b and b^* are in the column space $C(A)$, then $b + b^*$ is in $C(A)$.

26. If A is any 5 by 5 invertible matrix, then its column space is ________________. Why?

27. True or false (with a counterexample if false):

(a) The vectors b that are not in the column space $C(A)$ form a subspace.

(b) If $C(A)$ contains only the zero vector, then A is the zero matrix.

(c) The column space of $2A$ equals the column space of A.

(d) The column space of $A - I$ equals the column space of A (test this).

28. Construct a 3 by 3 matrix whose column space contains $(1, 1, 0)$ and $(1, 0, 1)$ but not $(1, 1, 1)$. Construct a 3 by 3 matrix whose column space is only a line.

29. If the 9 by 12 system $Ax = b$ is solvable for every b, then $C(A) =$ ________________.

Challenge Problems

30. Suppose S and T are two subspaces of a vector space V.

(a) Definition: The sum $S + T$ contains all sums $s + t$ of a vector s in S and a vector t in T. Show that $S + T$ satisfies the requirements (addition and scalar multiplication) for a vector space.

(b) If S and T are lines in $\mathbb{R}^n$, what is the difference between $S + T$ and $S \cup T$? That union contains all vectors from S or T or both. Explain this statement: *The span of* $S \cup T$ *is* $S + T$. (Section 3.5 returns to this word "span".)

31. If S is the column space of A and T is $C(B)$, then $S + T$ is the column space of what matrix M? The columns of A and B and M are all in $\mathbb{R}^m$. (I don't think $A + B$ is always correct M.)

32. Show that the matrices A and $\begin{bmatrix} A & AB \end{bmatrix}$ (with extra columns) have the same column space. But find a square matrix with $C(A^2)$ smaller than $C(A)$.
Important point: An n by n matrix has $C(A) = \mathbb{R}^n$ exactly when A is an ____________ matrix.

3.2 The Nullspace of A : Solving $Ax = 0$ and $Rx = 0$

Challenge Problems

3.3 The Complete Solution to $Ax = b$

Challenge Problems

3.4 Independence, Basis and Dimension

Challenge Problems

3.5 Dimensions of the Four Subspaces

Challenge Problems

4 Chapter 4: Orthogonality

4.1 Orthogonality of the Four Subspaces

Challenge Problems

4.2 Projections

Challenge Problems

4.3 Least Squares Approximations

Challenge Problems

4.4 Orthonormal Bases and Gram-Schmidt

Challenge Problems

5 Chapter 5: Determinants

5.1 The Properties of Determinants

Challenge Problems

5.2 Permutations and Cofactors

Challenge Problems

5.3 Cramer's Rule, Inverses, and Volumes

Challenge Problems

6 Chapter 6: Eigenvalues and Eigenvectors

6.1 Introduction to Eigenvalues

Challenge Problems

6.2 Diagonalizing a Matrix

Challenge Problems

6.3 Systems of Differential Equations

Challenge Problems

6.4 Symmetric Matrices

Challenge Problems

6.5 Positive Definite Matrices

Challenge Problems

7 Chapter 7: The Singular Value Decomposition (SVD)

7.1 Image Processing by Linear Algebra

Challenge Problems

7.2 Bases and Matrices in the SVD

Challenge Problems

7.3 Principal Component Analysis (PCA by the SVD)

Challenge Problems

7.4 The Geometry of the SVD

Challenge Problems